

DSC640 Week - Term Milestone 1

December 15, 2025

1 National Database of Childcare Prices - Comprehensive Data Analysis

This notebook provides a detailed analysis of childcare prices across the United States, exploring variations by geography, age groups, and time periods.

1.1 1. Load and Explore the Dataset

Import necessary libraries and load the data to understand its structure.

```
[1]: # Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

# Set display options
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 100)
warnings.filterwarnings('ignore')

# Set visualization style
sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (12, 6)

[4]: # Load the dataset
file_path = './nationaldatabaseofchildcareprices.xlsx'
df = pd.read_excel(file_path)

print(f"Dataset loaded successfully!")
print(f"Shape: {df.shape}")

Dataset loaded successfully!
Shape: (34567, 227)

[5]: # Display first few rows
df.head(10)
```

```

[5]: State_Name State_Abbreviation County_Name County_FIPS_Code StudyYear \
0 Alabama AL Autauga County 1001 2008
1 Alabama AL Autauga County 1001 2009
2 Alabama AL Autauga County 1001 2010
3 Alabama AL Autauga County 1001 2011
4 Alabama AL Autauga County 1001 2012
5 Alabama AL Autauga County 1001 2013
6 Alabama AL Autauga County 1001 2014
7 Alabama AL Autauga County 1001 2015
8 Alabama AL Autauga County 1001 2016
9 Alabama AL Autauga County 1001 2017

```

```

UNR_16 FUNR_16 MUNR_16 UNR_20to64 FUNR_20to64 MUNR_20to64 \
0 5.42 4.41 6.32 4.6 3.5 5.6
1 5.93 5.72 6.11 4.8 4.6 5.0
2 6.21 5.57 6.78 5.1 4.6 5.6
3 7.55 8.13 7.03 6.2 6.3 6.1
4 8.60 8.88 8.29 6.7 6.4 7.0
5 9.39 10.31 8.56 7.3 7.6 7.0
6 8.50 9.18 7.95 6.8 6.8 6.8
7 7.64 8.45 6.91 5.9 6.1 5.9
8 5.59 6.27 4.99 4.4 4.6 4.3
9 5.21 5.84 4.64 4.6 4.9 4.3

```

```

FLFPR_20to64 FLFPR_20to64_Under6 FLFPR_20to64_6to17 \
0 68.9 66.9 79.59
1 70.8 63.7 78.41
2 71.3 67.0 78.15
3 70.2 66.5 77.62
4 70.6 67.1 76.31
5 70.7 67.5 75.91
6 69.9 65.2 75.13
7 68.8 66.8 74.60
8 68.3 67.4 72.60
9 67.9 63.9 77.80

```

```

FLFPR_20to64_Under6_6to17 MLFPR_20to64 PR_F PR_P MHI ME \
0 60.81 84.0 8.5 11.5 50837.0 28444.0
1 59.91 86.2 7.5 10.3 51463.0 29648.0
2 59.71 85.8 7.5 10.6 53255.0 29949.0
3 59.31 85.7 7.4 10.9 53899.0 30861.0
4 58.30 85.7 7.4 11.6 53773.0 31207.0
5 58.00 85.0 8.3 12.1 53682.0 30868.0
6 57.40 84.2 9.1 12.8 52475.0 30210.0
7 57.00 82.5 9.3 12.9 51281.0 30760.0
8 53.30 81.9 9.4 12.3 53099.0 31259.0
9 53.30 82.0 10.9 13.7 55317.0 31891.0

```

	FME	MME	MHI_2018	ME_2018	FME_2018	MME_2018	TotalPop	\
0	21875.0	36032.0	58462.55	32710.60	25156.25	41436.80	49744	
1	22951.0	37492.0	60211.71	34688.16	26852.67	43865.64	49584	
2	23613.0	39789.0	61775.80	34740.84	27391.08	46155.24	53155	
3	23864.0	40476.0	60366.88	34564.32	26727.68	45333.12	53944	
4	25425.0	40251.0	59150.30	34327.70	27967.50	44276.10	54590	
5	25257.0	38889.0	57976.56	33337.44	27277.56	42000.12	54907	
6	23318.0	38189.0	55623.50	32022.60	24717.08	40480.34	55136	
7	22437.0	39069.0	54357.86	32605.60	23783.22	41413.14	55221	
8	24285.0	40192.0	55753.95	32821.95	25499.25	42201.60	55049	
9	24832.0	40866.0	56976.51	32847.73	25576.96	42091.98	55036	

	OneRace	OneRace_W	OneRace_B	OneRace_I	OneRace_A	OneRace_H	\
0	98.1	78.9	17.7	0.4	0.4	0.0	
1	98.6	79.1	17.9	0.4	0.6	0.0	
2	98.5	79.1	17.9	0.3	0.7	0.0	
3	98.5	78.9	18.1	0.2	0.7	0.0	
4	98.5	78.9	18.1	0.3	0.8	0.0	
5	98.6	78.3	18.4	0.3	1.0	0.0	
6	98.7	78.0	18.6	0.4	0.9	0.0	
7	98.4	77.4	18.7	0.4	1.0	0.0	
8	98.0	76.9	18.5	0.4	0.8	0.1	
9	98.3	76.6	19.1	0.3	0.9	0.1	

	OneRace_Other	TwoRaces	Hispanic	Households	H_Under6_BothWork	\
0	0.7	1.9	1.8	18373	1543	
1	0.7	1.4	2.0	18288	1475	
2	0.6	1.5	2.3	19718	1569	
3	0.5	1.5	2.4	19998	1695	
4	0.4	1.5	2.4	19934	1714	
5	0.7	1.4	2.5	20071	1532	
6	0.7	1.3	2.5	20304	1557	
7	0.9	1.6	2.6	20396	1306	
8	1.4	2.0	2.6	20800	1321	
9	1.3	1.7	2.7	21054	1243	

	H_Under6_FWork	H_Under6_MWork	H_Under6_SingleM	H_6to17_BothWork	\
0	970	22	995.0	4900	
1	964	16	1099.0	5028	
2	1009	16	1110.0	5472	
3	1060	106	1030.0	5065	
4	938	120	1095.0	4608	
5	880	161	1160.0	4238	
6	1191	159	954.0	4056	
7	1258	211	883.0	4030	
8	1361	109	731.0	4265	

9 1301 102 843.0 4557

	H_6to17_Fwork	H_6to17_Mwork	H_6to17_SingleM	EMP_M	MEMP_M	FEMP_M	\
0	1308	114	1966.0	27.40	24.41	30.68	
1	1519	92	2305.0	29.54	26.07	33.40	
2	1541	113	2377.0	29.33	25.94	33.06	
3	1965	246	2299.0	31.17	26.97	35.96	
4	1963	284	2644.0	32.13	28.59	36.09	
5	1978	354	2522.0	31.74	27.44	36.61	
6	2073	373	2269.0	32.32	28.13	37.05	
7	2086	551	2104.0	33.23	28.26	38.86	
8	1793	322	2168.0	33.82	29.78	38.39	
9	1503	341	2046.0	35.31	31.15	40.05	

	EMP_Service	MEMP_Service	FEMP_Service	EMP_Sales	MEMP_Sales	FEMP_Sales	\
0	17.06	15.53	18.75	29.11	15.97	43.52	
1	15.81	14.16	17.64	28.75	17.51	41.25	
2	16.92	15.09	18.93	29.07	17.82	41.43	
3	16.18	14.21	18.42	27.56	17.74	38.76	
4	16.09	14.71	17.63	28.39	17.79	40.26	
5	16.72	13.92	19.89	27.22	17.38	38.36	
6	16.81	13.78	20.24	25.42	15.99	36.07	
7	16.97	14.39	19.90	24.16	15.98	33.43	
8	17.48	16.05	19.11	24.19	15.33	34.21	
9	18.00	15.95	20.33	23.20	16.04	31.36	

	EMP_N	MEMP_N	FEMP_N	EMP_P	MEMP_P	FEMP_P	iUNR_16	iFUNR_16	iMUNR_16	\
0	13.21	22.54	2.99	13.22	21.55	4.07	0	0	0	
1	11.89	20.30	2.52	14.02	21.96	5.19	0	0	0	
2	11.57	19.86	2.45	13.11	21.28	4.13	0	0	0	
3	10.72	18.28	2.09	14.38	22.80	4.77	0	0	0	
4	9.02	16.03	1.19	14.37	22.88	4.84	0	0	0	
5	9.27	16.79	0.77	15.04	24.48	4.36	0	0	0	
6	9.38	17.16	0.58	16.08	24.94	6.07	0	0	0	
7	8.57	16.08	0.07	17.06	25.30	7.75	0	0	0	
8	7.85	14.70	0.10	16.66	24.14	8.19	0	0	0	
9	8.11	15.13	0.09	15.39	21.72	8.17	0	0	0	

	iUNR_20to64	iFUNR_20to64	iMUNR_20to64	iFLFPR_20to64	\
0	0	0	0	0	
1	0	0	0	0	
2	0	0	0	0	
3	0	0	0	0	
4	0	0	0	0	
5	0	0	0	0	
6	0	0	0	0	
7	0	0	0	0	

8	0	0	0	0
9	0	0	0	0

	iFLFPR_20to64_Under6	iFLFPR_20to64_6to17	iFLFPR_20to64_Under6_6to17	\
0	0	1	1	
1	0	1	1	
2	0	1	1	
3	0	1	1	
4	0	1	1	
5	0	1	1	
6	0	1	1	
7	0	0	0	
8	0	0	0	
9	0	0	0	

	iMLFPR_20to64	iPR_F	iPR_P	iMHI	iME	iFME	iMME	iMHI_2018	iME_2018	\
0	0	0	0	0	0	0	0	1	1	
1	0	0	0	0	0	0	0	1	1	
2	0	0	0	0	0	0	0	1	1	
3	0	0	0	0	0	0	0	1	1	
4	0	0	0	0	0	0	0	1	1	
5	0	0	0	0	0	0	0	1	1	
6	0	0	0	0	0	0	0	1	1	
7	0	0	0	0	0	0	0	1	1	
8	0	0	0	0	0	0	0	1	1	
9	0	0	0	0	0	0	0	1	1	

	iFME_2018	iMME_2018	iTotalPop	iOneRace	iOneRace_W	iOneRace_B	\
0	1	1	0	0	0	0	
1	1	1	0	0	0	0	
2	1	1	0	0	0	0	
3	1	1	0	0	0	0	
4	1	1	0	0	0	0	
5	1	1	0	0	0	0	
6	1	1	0	0	0	0	
7	1	1	0	0	0	0	
8	1	1	0	0	0	0	
9	1	1	0	0	0	0	

	iOneRace_I	iOneRace_A	iOneRace_H	iOneRace_Other	iTwoRaces	iHispanic	\
0	0	0	0	0	0	0	
1	0	0	0	0	0	0	
2	0	0	0	0	0	0	
3	0	0	0	0	0	0	
4	0	0	0	0	0	0	
5	0	0	0	0	0	0	
6	0	0	0	0	0	0	

7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0

	iHouseholds	iH_Under6_BothWork	iH_Under6_FWork	iH_Under6_MWork	\
0	0	0	0	0	
1	0	0	0	0	
2	0	0	0	0	
3	0	0	0	0	
4	0	0	0	0	
5	0	0	0	0	
6	0	0	0	0	
7	0	0	0	0	
8	0	0	0	0	
9	0	0	0	0	

	iH_Under6_SingleM	iH_6to17_BothWork	iH_6to17_Fwork	iH_6to17_Mwork	\
0	0	0	0	0	
1	0	0	0	0	
2	0	0	0	0	
3	0	0	0	0	
4	0	0	0	0	
5	0	0	0	0	
6	0	0	0	0	
7	0	0	0	0	
8	0	0	0	0	
9	0	0	0	0	

	iH_6to17_SingleM	iEMP_M	iMEMP_M	iFEMP_M	iEMP_Service	iMEMP_Service	\
0	0	1	1	1	1	1	
1	0	1	1	1	1	1	
2	0	1	1	1	1	1	
3	0	1	1	1	1	1	
4	0	1	1	1	1	1	
5	0	1	1	1	1	1	
6	0	1	1	1	1	1	
7	0	1	1	1	1	1	
8	0	1	1	1	1	1	
9	0	1	1	1	1	1	

	iFEMP_Service	iEMP_Sales	iMEMP_Sales	iFEMP_Sales	iEMP_N	iMEMP_N	\
0	1	1	1	1	1	1	
1	1	1	1	1	1	1	
2	1	1	1	1	1	1	
3	1	1	1	1	1	1	
4	1	1	1	1	1	1	
5	1	1	1	1	1	1	

6	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1

	iFEMP_N	iEMP_P	iMEMP_P	iFEMP_P	MCBto5	MC6to11	MC12to17	MC18to23	\
0	1	1	1	1	104.95	104.95	104.95	104.95	
1	1	1	1	1	105.11	105.11	105.11	105.11	
2	1	1	1	1	105.28	105.28	105.28	105.28	
3	1	1	1	1	105.45	105.45	105.45	105.45	
4	1	1	1	1	105.61	105.61	105.61	105.61	
5	1	1	1	1	105.78	105.78	105.78	105.78	
6	1	1	1	1	105.95	105.95	105.95	105.95	
7	1	1	1	1	109.56	109.56	109.56	109.56	
8	1	1	1	1	113.18	113.18	113.18	113.18	
9	1	1	1	1	116.79	116.79	116.79	116.79	

	MC24to29	MC30to35	MC36to41	MC42to47	MC48to53	MC54toSA	MCSA	\
0	104.95	85.92	85.92	85.92	85.92	85.92	80.92	
1	105.11	87.59	87.59	87.59	87.59	87.59	83.42	
2	105.28	89.26	89.26	89.26	89.26	89.26	85.92	
3	105.45	90.93	90.93	90.93	90.93	90.93	88.43	
4	105.61	92.60	92.60	92.60	92.60	92.60	90.93	
5	105.78	94.27	94.27	94.27	94.27	94.27	93.43	
6	105.95	95.94	95.94	95.94	95.94	95.94	95.94	
7	109.56	97.33	97.33	97.33	97.33	97.33	95.94	
8	113.18	98.72	98.72	98.72	98.72	98.72	95.94	
9	116.79	100.11	100.11	100.11	100.11	100.11	95.94	

	MFCCBto5	MFCC6to11	MFCC12to17	MFCC18to23	MFCC24to29	MFCC30to35	\
0	83.45	83.45	83.45	83.45	83.45	81.40	
1	87.39	87.39	87.39	87.39	87.39	85.68	
2	91.33	91.33	91.33	91.33	91.33	89.96	
3	95.28	95.28	95.28	95.28	95.28	94.25	
4	99.22	99.22	99.22	99.22	99.22	98.53	
5	103.16	103.16	103.16	103.16	103.16	102.82	
6	107.10	107.10	107.10	107.10	107.10	107.10	
7	107.10	107.10	107.10	107.10	107.10	106.81	
8	107.10	107.10	107.10	107.10	107.10	106.53	
9	107.10	107.10	107.10	107.10	107.10	106.24	

	MFCC36to41	MFCC42to47	MFCC48to53	MFCC54toSA	MFCCSA	_75CBto5	\
0	81.40	81.40	81.40	81.40	81.40	125.80	
1	85.68	85.68	85.68	85.68	85.68	126.00	
2	89.96	89.96	89.96	89.96	89.96	126.20	
3	94.25	94.25	94.25	94.25	94.25	126.40	
4	98.53	98.53	98.53	98.53	98.53	126.60	

5	102.82	102.82	102.82	102.82	102.82	126.80
6	107.10	107.10	107.10	107.10	107.10	127.00
7	106.81	106.81	106.81	106.81	106.53	131.33
8	106.53	106.53	106.53	106.53	105.96	135.67
9	106.24	106.24	106.24	106.24	105.39	140.00

	_75C6to11	_75C12to17	_75C18to23	_75C24to29	_75C30to35	_75C36to41 \
0	125.80	125.80	125.80	125.80	103.00	103.00
1	126.00	126.00	126.00	126.00	105.00	105.00
2	126.20	126.20	126.20	126.20	107.00	107.00
3	126.40	126.40	126.40	126.40	109.00	109.00
4	126.60	126.60	126.60	126.60	111.00	111.00
5	126.80	126.80	126.80	126.80	113.00	113.00
6	127.00	127.00	127.00	127.00	115.00	115.00
7	131.33	131.33	131.33	131.33	116.67	116.67
8	135.67	135.67	135.67	135.67	118.33	118.33
9	140.00	140.00	140.00	140.00	120.00	120.00

	_75C42to47	_75C48to53	_75C54toSA	_75CSA	_75FCCBto5	_75FCC6to11 \
0	103.00	103.00	103.00	97.0	97.4	97.4
1	105.00	105.00	105.00	100.0	102.0	102.0
2	107.00	107.00	107.00	103.0	106.6	106.6
3	109.00	109.00	109.00	106.0	111.2	111.2
4	111.00	111.00	111.00	109.0	115.8	115.8
5	113.00	113.00	113.00	112.0	120.4	120.4
6	115.00	115.00	115.00	115.0	125.0	125.0
7	116.67	116.67	116.67	115.0	125.0	125.0
8	118.33	118.33	118.33	115.0	125.0	125.0
9	120.00	120.00	120.00	115.0	125.0	125.0

	_75FCC12to17	_75FCC18to23	_75FCC24to29	_75FCC30to35	_75FCC36to41 \
0	97.4	97.4	97.4	95.00	95.00
1	102.0	102.0	102.0	100.00	100.00
2	106.6	106.6	106.6	105.00	105.00
3	111.2	111.2	111.2	110.00	110.00
4	115.8	115.8	115.8	115.00	115.00
5	120.4	120.4	120.4	120.00	120.00
6	125.0	125.0	125.0	125.00	125.00
7	125.0	125.0	125.0	124.67	124.67
8	125.0	125.0	125.0	124.33	124.33
9	125.0	125.0	125.0	124.00	124.00

	_75FCC42to47	_75FCC48to53	_75FCC54toSA	_75FCCSA	iMCBto5	iMC6to11 \
0	95.00	95.00	95.00	95.00	11.0	11.0
1	100.00	100.00	100.00	100.00	10.0	10.0
2	105.00	105.00	105.00	105.00	11.0	11.0
3	110.00	110.00	110.00	110.00	11.0	11.0

4	115.00	115.00	115.00	115.00	11.0	11.0
5	120.00	120.00	120.00	120.00	11.0	11.0
6	125.00	125.00	125.00	125.00	10.0	10.0
7	124.67	124.67	124.67	124.33	11.0	11.0
8	124.33	124.33	124.33	123.67	11.0	11.0
9	124.00	124.00	124.00	123.00	10.0	10.0

	iMC12to17	iMC18to23	iMC24to29	iMC30to35	iMC36to41	iMC42to47	\
0	11.0	11.0	11.0	11.0	11.0	11.0	
1	10.0	10.0	10.0	10.0	10.0	10.0	
2	11.0	11.0	11.0	11.0	11.0	11.0	
3	11.0	11.0	11.0	11.0	11.0	11.0	
4	11.0	11.0	11.0	11.0	11.0	11.0	
5	11.0	11.0	11.0	11.0	11.0	11.0	
6	10.0	10.0	10.0	10.0	10.0	10.0	
7	11.0	11.0	11.0	11.0	11.0	11.0	
8	11.0	11.0	11.0	11.0	11.0	11.0	
9	10.0	10.0	10.0	10.0	10.0	10.0	

	iMC48to53	iMC54toSA	iMCSA	iMFCCBto5	iMFCC6to11	iMFCC12to17	\
0	11.0	11.0	11.0	11.0	11.0	11.0	
1	10.0	10.0	10.0	10.0	10.0	10.0	
2	11.0	11.0	11.0	11.0	11.0	11.0	
3	11.0	11.0	11.0	11.0	11.0	11.0	
4	11.0	11.0	11.0	11.0	11.0	11.0	
5	11.0	11.0	11.0	11.0	11.0	11.0	
6	10.0	10.0	10.0	10.0	10.0	10.0	
7	11.0	11.0	11.0	11.0	11.0	11.0	
8	11.0	11.0	11.0	11.0	11.0	11.0	
9	10.0	10.0	10.0	10.0	10.0	10.0	

	iMFCC18to23	iMFCC24to29	iMFCC30to35	iMFCC36to41	iMFCC42to47	\
0	11.0	11.0	11.0	11.0	11.0	
1	10.0	10.0	10.0	10.0	10.0	
2	11.0	11.0	11.0	11.0	11.0	
3	11.0	11.0	11.0	11.0	11.0	
4	11.0	11.0	11.0	11.0	11.0	
5	11.0	11.0	11.0	11.0	11.0	
6	10.0	10.0	10.0	10.0	10.0	
7	11.0	11.0	11.0	11.0	11.0	
8	11.0	11.0	11.0	11.0	11.0	
9	10.0	10.0	10.0	10.0	10.0	

	iMFCC48to53	iMFCC54toSA	iMFCCSA	i_75CBto5	i_75C6to11	i_75C12to17	\
0	11.0	11.0	11.0	1.0	1.0	1.0	
1	10.0	10.0	10.0	0.0	0.0	0.0	
2	11.0	11.0	11.0	1.0	1.0	1.0	

3	11.0	11.0	11.0	1.0	1.0	1.0
4	11.0	11.0	11.0	1.0	1.0	1.0
5	11.0	11.0	11.0	1.0	1.0	1.0
6	10.0	10.0	10.0	0.0	0.0	0.0
7	11.0	11.0	11.0	1.0	1.0	1.0
8	11.0	11.0	11.0	1.0	1.0	1.0
9	10.0	10.0	10.0	0.0	0.0	0.0

	i_75C18to23	i_75C24to29	i_75C30to35	i_75C36to41	i_75C42to47	\
0	1.0	1.0	1.0	1.0	1.0	
1	0.0	0.0	0.0	0.0	0.0	
2	1.0	1.0	1.0	1.0	1.0	
3	1.0	1.0	1.0	1.0	1.0	
4	1.0	1.0	1.0	1.0	1.0	
5	1.0	1.0	1.0	1.0	1.0	
6	0.0	0.0	0.0	0.0	0.0	
7	1.0	1.0	1.0	1.0	1.0	
8	1.0	1.0	1.0	1.0	1.0	
9	0.0	0.0	0.0	0.0	0.0	

	i_75C48to53	i_75C54toSA	i_75CSA	i_75FCCBto5	i_75FCC6to11	\
0	1.0	1.0	1.0	1.0	1.0	
1	0.0	0.0	0.0	0.0	0.0	
2	1.0	1.0	1.0	1.0	1.0	
3	1.0	1.0	1.0	1.0	1.0	
4	1.0	1.0	1.0	1.0	1.0	
5	1.0	1.0	1.0	1.0	1.0	
6	0.0	0.0	0.0	0.0	0.0	
7	1.0	1.0	1.0	1.0	1.0	
8	1.0	1.0	1.0	1.0	1.0	
9	0.0	0.0	0.0	0.0	0.0	

	i_75FCC12to17	i_75FCC18to23	i_75FCC24to29	i_75FCC30to35	i_75FCC36to41	\
0	1.0	1.0	1.0	1.0	1.0	
1	0.0	0.0	0.0	0.0	0.0	
2	1.0	1.0	1.0	1.0	1.0	
3	1.0	1.0	1.0	1.0	1.0	
4	1.0	1.0	1.0	1.0	1.0	
5	1.0	1.0	1.0	1.0	1.0	
6	0.0	0.0	0.0	0.0	0.0	
7	1.0	1.0	1.0	1.0	1.0	
8	1.0	1.0	1.0	1.0	1.0	
9	0.0	0.0	0.0	0.0	0.0	

	i_75FCC42to47	i_75FCC48to53	i_75FCC54toSA	i_75FCCSA	MCInfant	\
0	1.0	1.0	1.0	1.0	104.95	
1	0.0	0.0	0.0	0.0	105.11	

2	1.0	1.0	1.0	1.0	105.28
3	1.0	1.0	1.0	1.0	105.45
4	1.0	1.0	1.0	1.0	105.61
5	1.0	1.0	1.0	1.0	105.78
6	0.0	0.0	0.0	0.0	105.95
7	1.0	1.0	1.0	1.0	109.56
8	1.0	1.0	1.0	1.0	113.18
9	0.0	0.0	0.0	0.0	116.79

	MCInfant_flag	MCToddler	MCToddler_flag	MCPreschool	MCPreschool_flag \
0	1.0	104.95	3.0	85.92	1.0
1	1.0	105.11	3.0	87.59	1.0
2	1.0	105.28	3.0	89.26	1.0
3	1.0	105.45	3.0	90.93	1.0
4	1.0	105.61	3.0	92.60	1.0
5	1.0	105.78	3.0	94.27	1.0
6	1.0	105.95	3.0	95.94	1.0
7	1.0	109.56	3.0	97.33	1.0
8	1.0	113.18	3.0	98.72	1.0
9	1.0	116.79	3.0	100.11	1.0

	_75CInfant	_75CInfant_flag	_75CToddler	_75CToddler_flag	_75CPreschool \
0	125.80	1.0	125.80	3.0	103.00
1	126.00	1.0	126.00	3.0	105.00
2	126.20	1.0	126.20	3.0	107.00
3	126.40	1.0	126.40	3.0	109.00
4	126.60	1.0	126.60	3.0	111.00
5	126.80	1.0	126.80	3.0	113.00
6	127.00	1.0	127.00	3.0	115.00
7	131.33	1.0	131.33	3.0	116.67
8	135.67	1.0	135.67	3.0	118.33
9	140.00	1.0	140.00	3.0	120.00

	_75CPreschool_flag	MFCCInfant	MFCCInfant_flag	MFCCToddler \
0	1.0	83.45	1.0	83.45
1	1.0	87.39	1.0	87.39
2	1.0	91.33	1.0	91.33
3	1.0	95.28	1.0	95.28
4	1.0	99.22	1.0	99.22
5	1.0	103.16	1.0	103.16
6	1.0	107.10	1.0	107.10
7	1.0	107.10	1.0	107.10
8	1.0	107.10	1.0	107.10
9	1.0	107.10	1.0	107.10

	MFCCToddler_flag	MFCCPreschool	MFCCPreschool_flag	_75FCCInfant \
0	3.0	81.40	1.0	97.4

1	3.0	85.68	1.0	102.0
2	3.0	89.96	1.0	106.6
3	3.0	94.25	1.0	111.2
4	3.0	98.53	1.0	115.8
5	3.0	102.82	1.0	120.4
6	1.0	107.10	1.0	125.0
7	3.0	106.81	1.0	125.0
8	3.0	106.53	1.0	125.0
9	3.0	106.24	1.0	125.0

	_75FCCInfant_flag	_75FCCToddler	_75FCCToddler_flag	_75FCCPreschool \
0	1.0	97.4	3.0	95.00
1	1.0	102.0	3.0	100.00
2	1.0	106.6	3.0	105.00
3	1.0	111.2	3.0	110.00
4	1.0	115.8	3.0	115.00
5	1.0	120.4	3.0	120.00
6	1.0	125.0	1.0	125.00
7	1.0	125.0	3.0	124.67
8	1.0	125.0	3.0	124.33
9	1.0	125.0	3.0	124.00

	_75FCCPreschool_flag
0	1.0
1	1.0
2	1.0
3	1.0
4	1.0
5	1.0
6	1.0
7	1.0
8	1.0
9	1.0

```
[6]: # Display column names and data types
print("Column Names and Data Types:")
print("="*60)
df.dtypes
```

Column Names and Data Types:

=====

```
[6]: State_Name          object
State_Abbreviation      object
County_Name             object
County_FIPS_Code        int64
StudyYear               int64
```

...

```

_75FCCInfant_flag      float64
_75FCCToddler          float64
_75FCCToddler_flag     float64
_75FCCPreschool        float64
_75FCCPreschool_flag   float64
Length: 227, dtype: object

```

```

[7]: # Get basic information about the dataset
df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 34567 entries, 0 to 34566
Columns: 227 entries, State_Name to _75FCCPreschool_flag
dtypes: float64(159), int64(65), object(3)
memory usage: 59.9+ MB

```

1.2 2. Data Cleaning and Preprocessing

Check for data quality issues and prepare the dataset for analysis.

```

[9]: # Check for missing values
missing_data = df.isnull().sum()
missing_percent = (missing_data / len(df)) * 100

missing_df = pd.DataFrame({
    'Missing_Count': missing_data,
    'Percentage': missing_percent
}).sort_values('Percentage', ascending=False)

print("Missing Values Summary:")
print("="*60)
print(missing_df[missing_df['Missing_Count'] > 0])

```

Missing Values Summary:

```

=====

```

	Missing_Count	Percentage
iMFCC12to17	11204	32.412416
iMFCCBto5	11204	32.412416
iMFCC54toSA	11204	32.412416
iMFCC48to53	11204	32.412416
iMFCC42to47	11204	32.412416
...	...	...
_75CPreschool_flag	10974	31.747042
_75CSA	10974	31.747042
MC6to11	10974	31.747042
H_6to17_SingleM	2	0.005786
H_Under6_SingleM	2	0.005786

```

[114 rows x 2 columns]

```

```
[10]: # Check for duplicate rows
duplicates = df.duplicated().sum()
print(f"Number of duplicate rows: {duplicates}")

# Display basic statistics about the dataset
print(f"\nDataset Overview:")
print(f"Total rows: {len(df):,}")
print(f"Total columns: {len(df.columns)}")
```

Number of duplicate rows: 0

Dataset Overview:
Total rows: 34,567
Total columns: 227

```
[12]: # Identify categorical columns
categorical_columns = df.select_dtypes(include=['object']).columns.tolist()
print(f"Categorical columns ({len(categorical_columns)}):")
for col in categorical_columns:
    print(f"    - {col}")
```

Categorical columns (3):
- State_Name
- State_Abbreviation
- County_Name

```
[13]: # Create a copy for analysis to preserve original data
df_clean = df.copy()

# Display unique values for key categorical columns (first few)
print("Unique values in key categorical columns:")
print("="*60)
for col in categorical_columns[:5]:
    n_unique = df_clean[col].nunique()
    print(f"\n{col}: {n_unique} unique values")
    if n_unique <= 10:
        print(df_clean[col].unique())
```

Unique values in key categorical columns:
=====

State_Name: 51 unique values

State_Abbreviation: 51 unique values

County_Name: 1879 unique values

1.3 3. Statistical Summary and Descriptive Analysis

Generate summary statistics for numerical columns and analyze distributions.

```
[14]: # Generate descriptive statistics for all numeric columns
df_clean.describe()
```

```
[14]:
```

	County_FIPS_Code	StudyYear	UNR_16	FUNR_16	\
count	34567.000000	34567.000000	34567.000000	34567.000000	
mean	30388.132786	2012.999711	7.465902	7.02902	
std	15161.015383	3.162232	3.538619	3.56342	
min	1001.000000	2008.000000	0.000000	0.00000	
25%	18177.000000	2010.000000	5.100000	4.64000	
50%	29177.000000	2013.000000	7.050000	6.59000	
75%	45081.000000	2016.000000	9.350000	8.88000	
max	56045.000000	2018.000000	36.110000	38.24000	

	MUNR_16	UNR_20to64	FUNR_20to64	MUNR_20to64	FLFPR_20to64	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	7.860291	6.900073	6.482007	7.275457	70.086125	
std	4.037657	3.446199	3.477956	3.990758	7.696499	
min	0.000000	0.000000	0.000000	0.000000	33.600000	
25%	5.200000	4.600000	4.200000	4.700000	65.100000	
50%	7.390000	6.500000	6.000000	6.800000	70.600000	
75%	9.920000	8.700000	8.250000	9.200000	75.500000	
max	39.740000	33.900000	44.500000	45.500000	100.000000	

	FLFPR_20to64_Under6	FLFPR_20to64_6to17	FLFPR_20to64_Under6_6to17	\
count	34567.000000	34567.000000	34567.000000	
mean	68.821409	78.824106	66.940759	
std	11.758088	8.529813	13.285061	
min	0.000000	0.000000	0.000000	
25%	62.600000	74.400000	59.800000	
50%	69.600000	79.450000	67.660000	
75%	76.100000	84.225000	75.200000	
max	100.000000	100.000000	100.000000	

	MLFPR_20to64	PR_F	PR_P	MHI	ME	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	78.637814	11.739125	16.038131	46432.001931	27075.054081	
std	10.802983	5.681003	6.511816	12457.050093	5483.484930	
min	8.200000	0.000000	0.000000	18528.560000	4417.000000	
25%	74.200000	7.700000	11.400000	38250.500000	23636.000000	
50%	81.000000	10.900000	15.200000	44438.000000	26366.000000	
75%	85.900000	14.600000	19.500000	52078.390000	30022.500000	
max	100.000000	52.100000	55.100000	136268.000000	70565.000000	

	FME	MME	MHI_2018	ME_2018	FME_2018	\
--	-----	-----	----------	---------	----------	---

count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	21625.047495	33134.232570	50446.777363	29404.295539	23475.169710
std	4716.600817	7240.296013	13279.833788	5715.192737	4903.836942
min	4290.000000	3784.000000	19841.920000	4947.040000	5019.300000
25%	18721.000000	28729.000000	41839.215000	25955.055000	20613.100000
50%	21089.000000	32059.000000	48505.600000	28653.920000	22854.700000
75%	23919.000000	36543.500000	56201.450000	32005.820000	25634.665000
max	62919.000000	93000.000000	136268.000000	70565.000000	68600.000000

	MME_2018	TotalPop	OneRace	OneRace_W	OneRace_B \
count	34567.000000	3.456700e+04	34567.000000	34567.000000	34567.000000
mean	35997.433918	9.914222e+04	97.922863	83.636247	8.971213
std	7643.855532	3.177786e+05	1.952523	16.683515	14.510260
min	4238.080000	4.100000e+01	59.500000	3.100000	0.000000
25%	31447.505000	1.101450e+04	97.500000	77.000000	0.600000
50%	35103.000000	2.571100e+04	98.400000	90.200000	2.100000
75%	39503.335000	6.670350e+04	98.900000	95.600000	10.200000
max	98580.000000	1.010572e+07	100.000000	100.000000	87.400000

	OneRace_I	OneRace_A	OneRace_H	OneRace_Other	TwoRaces \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	1.895634	1.216209	0.082101	2.117914	2.077146
std	7.517786	2.633768	0.593602	3.779858	1.952520
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.200000	0.200000	0.000000	0.300000	1.100000
50%	0.300000	0.500000	0.000000	0.800000	1.600000
75%	0.800000	1.200000	0.100000	2.200000	2.500000
max	95.700000	52.200000	35.300000	58.800000	40.500000

	Hispanic	Households	H_Under6_BothWork	H_Under6_FWork \
count	34567.000000	3.456700e+04	34567.000000	34567.000000
mean	8.401054	3.686299e+04	2791.610004	1821.628808
std	13.284933	1.116470e+05	8709.085062	6740.389344
min	0.000000	2.200000e+01	0.000000	0.000000
25%	1.700000	4.236000e+03	260.000000	146.000000
50%	3.400000	9.814000e+03	647.000000	387.000000
75%	8.400000	2.549350e+04	1786.000000	1099.000000
max	99.200000	3.306109e+06	264439.000000	246269.000000

	H_Under6_MWork	H_Under6_SingleM	H_6to17_BothWork	H_6to17_Fwork \
count	34567.000000	34565.000000	34567.000000	34567.000000
mean	141.109064	1992.249964	6480.568548	2874.374374
std	472.557239	7009.934446	19294.344762	10901.264277
min	0.000000	0.000000	0.000000	0.000000
25%	5.000000	176.000000	685.000000	247.000000
50%	31.000000	489.000000	1662.000000	620.000000
75%	103.000000	1298.000000	4469.500000	1767.000000

max	15396.000000	220530.000000	590730.000000	391225.000000
-----	--------------	---------------	---------------	---------------

	H_6to17_Mwork	H_6to17_SingleM	EMP_M	MEMP_M \
count	34567.000000	34565.000000	34567.000000	34567.000000
mean	390.113056	3924.511558	30.720191	26.495400
std	1187.096922	13966.245194	6.503783	8.013214
min	0.000000	0.000000	0.000000	0.000000
25%	39.000000	351.000000	26.340000	20.945000
50%	116.000000	907.000000	29.690000	25.190000
75%	306.000000	2421.000000	34.100000	30.720000
max	40499.000000	436538.000000	74.030000	76.890000

	FEMP_M	EMP_Service	MEMP_Service	FEMP_Service	EMP_Sales \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	35.509491	17.909716	13.122111	23.424303	22.244044
std	6.073988	3.602868	4.476415	4.696024	3.352621
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	31.560000	15.590000	10.270000	20.410000	20.150000
50%	34.940000	17.580000	12.840000	23.090000	22.410000
75%	38.870000	19.780000	15.450000	26.010000	24.540000
max	100.000000	46.420000	56.330000	62.780000	50.770000

	MEMP_Sales	FEMP_Sales	EMP_N	MEMP_N	FEMP_N \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	13.660359	32.062036	13.076182	23.265887	1.369745
std	3.761803	4.480591	4.257660	6.763236	1.430796
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	11.210000	29.390000	10.200000	18.680000	0.590000
50%	13.680000	32.170000	12.560000	22.700000	0.990000
75%	16.240000	34.800000	15.340000	27.210000	1.650000
max	49.060000	100.000000	60.000000	80.240000	27.270000

	EMP_P	MEMP_P	FEMP_P	iUNR_16	iFUNR_16 \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	16.049861	23.456238	7.634403	0.038881	0.038881
std	5.935141	7.956927	4.501044	0.193314	0.193314
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	11.640000	17.610000	4.410000	0.000000	0.000000
50%	15.590000	23.400000	6.630000	0.000000	0.000000
75%	19.830000	28.930000	9.920000	0.000000	0.000000
max	72.850000	87.500000	66.480000	1.000000	1.000000

	iMUNR_16	iUNR_20to64	iFUNR_20to64	iMUNR_20to64	iFLFPR_20to64 \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	0.038881	0.038881	0.038881	0.038881	0.038881
std	0.193314	0.193314	0.193314	0.193314	0.193314
min	0.000000	0.000000	0.000000	0.000000	0.000000

25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	0.000000	0.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

	iFLFPR_20to64_Under6	iFLFPR_20to64_6to17	iFLFPR_20to64_Under6_6to17	\
count	34567.000000	34567.000000	34567.000000	
mean	0.040530	0.636590	0.637082	
std	0.197201	0.480988	0.480849	
min	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	
50%	0.000000	1.000000	1.000000	
75%	0.000000	1.000000	1.000000	
max	1.000000	1.000000	1.000000	

	iMLFPR_20to64	iPR_F	iPR_P	iMHI	iME	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	0.038881	0.038245	0.038158	0.038245	0.038447	
std	0.193314	0.191789	0.191580	0.191789	0.192276	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.000000	0.000000	0.000000	
max	1.000000	1.000000	1.000000	1.000000	1.000000	

	iFME	iMME	iMHI_2018	iME_2018	iFME_2018	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	0.038476	0.038505	0.909133	0.909191	0.909104	
std	0.192345	0.192415	0.287424	0.287342	0.287465	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	1.000000	1.000000	1.000000	
50%	0.000000	0.000000	1.000000	1.000000	1.000000	
75%	0.000000	0.000000	1.000000	1.000000	1.000000	
max	1.000000	1.000000	1.000000	1.000000	1.000000	

	iMME_2018	iTotalPop	iOneRace	iOneRace_W	iOneRace_B	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	0.90922	0.038187	0.044956	0.044956	0.044956	
std	0.28730	0.191650	0.207211	0.207211	0.207211	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	0.000000	0.000000	0.000000	0.000000	
50%	1.000000	0.000000	0.000000	0.000000	0.000000	
75%	1.000000	0.000000	0.000000	0.000000	0.000000	
max	1.000000	1.000000	1.000000	1.000000	1.000000	

	iOneRace_I	iOneRace_A	iOneRace_H	iOneRace_Other	iTwoRaces	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	

mean	0.044956	0.044956	0.044956	0.044956	0.044956
std	0.207211	0.207211	0.207211	0.207211	0.207211
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	0.000000	0.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

	iHispanic	iHouseholds	iH_Under6_BothWork	iH_Under6_FWork	\
count	34567.000000	34567.000000	34567.000000	34567.000000	
mean	0.046808	0.038187	0.038447	0.038447	
std	0.211230	0.191650	0.192276	0.192276	
min	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.000000	0.000000	
max	1.000000	1.000000	1.000000	1.000000	

	iH_Under6_MWork	iH_Under6_SingleM	iH_6to17_BothWork	iH_6to17_Fwork	\
count	34567.000000	34567.000000	34567.000000	34567.000000	
mean	0.038447	0.040732	0.038447	0.038447	
std	0.192276	0.197673	0.192276	0.192276	
min	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.000000	0.000000	
max	1.000000	1.000000	1.000000	1.000000	

	iH_6to17_Mwork	iH_6to17_SingleM	iEMP_M	iMEMP_M	iFEMP_M	\
count	34567.000000	34567.000000	34567.0	34567.0	34567.0	
mean	0.038447	0.040732	1.0	1.0	1.0	
std	0.192276	0.197673	0.0	0.0	0.0	
min	0.000000	0.000000	1.0	1.0	1.0	
25%	0.000000	0.000000	1.0	1.0	1.0	
50%	0.000000	0.000000	1.0	1.0	1.0	
75%	0.000000	0.000000	1.0	1.0	1.0	
max	1.000000	1.000000	1.0	1.0	1.0	

	iEMP_Service	iMEMP_Service	iFEMP_Service	iEMP_Sales	iMEMP_Sales	\
count	34567.0	34567.0	34567.0	34567.0	34567.0	
mean	1.0	1.0	1.0	1.0	1.0	
std	0.0	0.0	0.0	0.0	0.0	
min	1.0	1.0	1.0	1.0	1.0	
25%	1.0	1.0	1.0	1.0	1.0	
50%	1.0	1.0	1.0	1.0	1.0	
75%	1.0	1.0	1.0	1.0	1.0	
max	1.0	1.0	1.0	1.0	1.0	

	iFEMP_Sales	iEMP_N	iMEMP_N	iFEMP_N	iEMP_P	iMEMP_P	iFEMP_P	\
count	34567.0	34567.0	34567.0	34567.0	34567.0	34567.0	34567.0	
mean	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
std	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
min	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
25%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
50%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
75%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
max	1.0	1.0	1.0	1.0	1.0	1.0	1.0	

	MCBto5	MC6to11	MC12to17	MC18to23	MC24to29	\
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000	
mean	145.930609	145.929977	142.769363	139.820997	130.482768	
std	53.722917	53.730359	52.498825	51.290259	43.775370	
min	27.730000	27.730000	27.730000	27.730000	21.540000	
25%	108.450000	108.450000	107.000000	104.070000	100.000000	
50%	134.400000	134.400000	131.150000	128.540000	120.990000	
75%	166.330000	166.330000	165.000000	161.080000	148.710000	
max	470.000000	470.000000	470.000000	462.330000	419.000000	

	MC30to35	MC36to41	MC42to47	MC48to53	MC54toSA	\
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000	
mean	128.298475	122.197682	122.197192	120.114848	120.115166	
std	42.444909	38.528236	38.528509	37.474593	37.474395	
min	21.540000	21.540000	21.540000	21.540000	21.540000	
25%	98.670000	95.880000	95.880000	94.790000	94.790000	
50%	120.000000	113.980000	113.980000	112.000000	112.000000	
75%	147.200000	139.280000	139.280000	136.500000	136.500000	
max	419.000000	385.000000	385.000000	385.000000	385.000000	

	MCSA	MFCCBto5	MFCC6to11	MFCC12to17	MFCC18to23	\
count	23593.000000	23383.000000	23383.000000	23383.000000	23383.000000	
mean	101.234253	113.252743	113.228623	111.982142	109.849199	
std	34.552888	32.946207	32.897397	32.423459	31.923247	
min	18.980000	40.040000	40.040000	43.080000	43.080000	
25%	78.650000	90.000000	90.000000	89.955000	87.110000	
50%	96.530000	106.000000	106.000000	105.100000	103.850000	
75%	119.380000	129.250000	129.250000	128.000000	125.000000	
max	375.400000	351.290000	351.290000	430.940000	430.940000	

	MFCC24to29	MFCC30to35	MFCC36to41	MFCC42to47	MFCC48to53	\
count	23383.000000	23383.000000	23383.000000	23383.000000	23383.000000	
mean	106.739308	106.301836	104.189913	104.163946	103.331517	
std	29.970529	29.477408	28.960359	28.956597	28.837612	
min	43.080000	43.080000	40.030000	40.030000	40.030000	
25%	85.000000	85.060000	83.985000	83.985000	83.030000	

50%	100.250000	100.000000	99.660000	99.650000	99.000000
75%	124.950000	124.500000	120.200000	120.200000	119.960000
max	376.320000	376.320000	331.340000	331.340000	331.340000

	MFCC54toSA	MFCCSA	_75CBto5	_75C6to11	_75C12to17	\
count	23383.000000	23383.000000	23593.000000	23593.000000	23593.000000	
mean	103.319273	92.523582	165.747724	165.723671	162.313312	
std	28.823258	27.669904	62.948905	62.923413	61.354370	
min	40.030000	22.000000	32.180000	32.180000	32.180000	
25%	83.030000	75.000000	123.450000	123.450000	120.180000	
50%	99.000000	88.180000	150.000000	150.000000	148.130000	
75%	119.960000	107.500000	185.000000	185.000000	182.600000	
max	331.340000	308.000000	515.000000	515.000000	515.000000	

	_75C18to23	_75C24to29	_75C30to35	_75C36to41	_75C42to47	\
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000	
mean	159.300156	149.211592	146.869878	140.109674	140.344320	
std	60.338314	52.230764	50.595279	46.157201	46.329759	
min	32.180000	25.000000	25.000000	25.000000	25.000000	
25%	119.000000	113.580000	111.670000	108.780000	108.780000	
50%	145.000000	137.400000	135.000000	129.230000	129.380000	
75%	180.000000	169.330000	167.320000	158.680000	159.540000	
max	492.210000	485.000000	485.000000	430.000000	430.000000	

	_75C48to53	_75C54toSA	_75CSA	_75FCCBto5	_75FCC6to11	\
count	23593.000000	23593.000000	23593.000000	23383.000000	23383.000000	
mean	138.134505	138.135380	118.607736	128.621556	128.646967	
std	45.395519	45.395017	40.732376	38.586132	38.599945	
min	25.000000	25.000000	22.030000	47.210000	47.210000	
25%	106.880000	106.880000	92.130000	100.040000	100.040000	
50%	127.500000	127.500000	112.500000	122.650000	122.800000	
75%	156.650000	156.650000	138.890000	146.950000	146.950000	
max	430.000000	430.000000	540.000000	410.000000	410.000000	

	_75FCC12to17	_75FCC18to23	_75FCC24to29	_75FCC30to35	_75FCC36to41	\
count	23383.000000	23383.000000	23383.000000	23383.000000	23383.000000	
mean	127.100881	124.452879	120.755554	120.081898	117.868539	
std	37.887589	37.178615	35.302723	34.840404	34.101515	
min	50.000000	50.000000	50.000000	50.000000	46.450000	
25%	100.000000	98.710000	95.595000	95.450000	95.000000	
50%	121.460000	120.000000	115.000000	115.000000	112.500000	
75%	146.000000	141.430000	136.270000	136.250000	132.760000	
max	502.970000	502.970000	439.220000	439.220000	386.720000	

	_75FCC42to47	_75FCC48to53	_75FCC54toSA	_75FCCSA	iMCBto5	\
count	23383.000000	23383.000000	23383.000000	23383.000000	23593.000000	
mean	117.868539	117.060965	117.060965	107.514791	538.150045	

std	34.101515	33.957578	33.957578	31.439791	747.570838
min	46.450000	46.450000	46.450000	39.950000	0.000000
25%	95.000000	94.225000	94.225000	86.250000	1.000000
50%	112.500000	111.670000	111.670000	100.000000	111.000000
75%	132.760000	131.540000	131.540000	125.000000	1001.000000
max	386.720000	386.720000	386.720000	341.220000	11001.000000

	iMC6to11	iMC12to17	iMC18to23	iMC24to29	iMC30to35	\
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000	
mean	538.150045	527.129827	527.553681	514.414191	514.414191	
std	747.570838	676.514314	679.934582	584.844914	584.844914	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	111.000000	111.000000	111.000000	111.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
max	11001.000000	11001.000000	11001.000000	11000.000000	11000.000000	

	iMC36to41	iMC42to47	iMC48to53	iMC54toSA	iMCSA	\
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000	
mean	509.751791	509.751791	509.751791	509.751791	593.674988	
std	544.014342	544.014342	544.014342	544.014342	1081.932707	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	111.000000	111.000000	111.000000	111.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
max	10001.000000	10001.000000	10001.000000	10001.000000	11101.000000	

	iMFCCBto5	iMFCC6to11	iMFCC12to17	iMFCC18to23	iMFCC24to29	\
count	23363.000000	23363.000000	23363.000000	23363.000000	23363.000000	
mean	535.735094	537.447203	529.742713	513.477678	501.492916	
std	783.397312	795.248421	745.691945	641.250142	547.793159	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	111.000000	111.000000	111.000000	111.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
max	11100.000000	11100.000000	11100.000000	11100.000000	11100.000000	

	iMFCC30to35	iMFCC36to41	iMFCC42to47	iMFCC48to53	iMFCC54toSA	\
count	23363.000000	23363.000000	23363.000000	23363.000000	23363.000000	
mean	501.492916	499.352780	499.352780	498.068698	498.068698	
std	547.793159	525.786117	525.786117	512.172349	512.172349	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	111.000000	111.000000	111.000000	111.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
max	11100.000000	11100.000000	11100.000000	11100.000000	11100.000000	

	iMFCCSA	i_75CBto5	i_75C6to11	i_75C12to17	i_75C18to23	\
count	23363.000000	23575.000000	23575.000000	23575.000000	23575.000000	
mean	559.704618	594.387869	594.387869	581.238346	581.662524	
std	920.348454	792.244081	792.244081	713.358381	716.573259	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	1000.000000	1000.000000	1000.000000	1000.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
max	11101.000000	11001.000000	11001.000000	11001.000000	11001.000000	

	i_75C24to29	i_75C30to35	i_75C36to41	i_75C42to47	i_75C48to53	\
count	23575.000000	23575.000000	23575.000000	23575.000000	23575.000000	
mean	564.695398	564.695398	571.482248	561.301972	561.301972	
std	594.180636	594.180636	639.646517	564.529253	564.529253	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
max	11000.000000	11000.000000	11000.000000	11000.000000	11000.000000	

	i_75C54toSA	i_75CSA	i_75FCCBto5	i_75FCC6to11	i_75FCC12to17	\
count	23575.000000	23575.000000	23383.000000	23383.000000	23383.000000	
mean	561.301972	666.498155	596.021811	597.732455	590.889877	
std	564.529253	1155.682656	786.101669	797.773319	755.200166	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
max	11000.000000	11100.000000	11100.000000	11100.000000	11100.000000	

	i_75FCC18to23	i_75FCC24to29	i_75FCC30to35	i_75FCC36to41	\
count	23383.000000	23383.000000	23383.000000	23383.000000	
mean	574.638755	562.236582	562.236582	566.513193	
std	653.899222	559.793463	559.793463	589.392174	
min	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	
50%	1000.000000	1000.000000	1000.000000	1000.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	
max	11100.000000	11100.000000	11100.000000	11100.000000	

	i_75FCC42to47	i_75FCC48to53	i_75FCC54toSA	i_75FCCSA	\
count	23383.000000	23383.000000	23383.000000	23383.000000	
mean	566.513193	566.513193	566.513193	663.164607	
std	589.392174	589.392174	589.392174	1101.122681	
min	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	
50%	1000.000000	1000.000000	1000.000000	1000.000000	

75%	1001.000000	1001.000000	1001.000000	1001.000000
max	11100.000000	11100.000000	11100.000000	11100.000000

	MCInfant	MCInfant_flag	MCToddler	MCToddler_flag \
count	23593.000000	23593.000000	23593.000000	23593.000000
mean	146.051770	1.899801	130.482768	1.315263
std	53.698566	0.831546	43.775370	0.728806
min	27.730000	1.000000	21.540000	1.000000
25%	108.750000	1.000000	100.000000	1.000000
50%	134.500000	2.000000	120.990000	1.000000
75%	166.330000	3.000000	148.710000	1.000000
max	470.000000	3.000000	419.000000	3.000000

	MCPreschool	MCPreschool_flag	_75CInfant	_75CInfant_flag \
count	23593.000000	23593.000000	23593.000000	23593.000000
mean	122.232852	1.367609	165.926607	1.926461
std	38.538323	0.774638	62.889701	0.837951
min	21.540000	1.000000	32.180000	1.000000
25%	95.880000	1.000000	123.700000	1.000000
50%	113.990000	1.000000	150.000000	2.000000
75%	139.300000	1.000000	185.000000	3.000000
max	385.000000	3.000000	515.000000	3.000000

	_75CToddler	_75CToddler_flag	_75CPreschool	_75CPreschool_flag \
count	23593.000000	23593.000000	23593.000000	23593.000000
mean	149.213983	1.307634	140.386184	1.370661
std	52.236542	0.721561	46.337349	0.774716
min	25.000000	1.000000	25.000000	1.000000
25%	113.580000	1.000000	108.900000	1.000000
50%	137.400000	1.000000	129.450000	1.000000
75%	169.330000	1.000000	159.580000	1.000000
max	485.000000	3.000000	430.000000	3.000000

	MFCCInfant	MFCCInfant_flag	MFCCToddler	MFCCToddler_flag \
count	23383.000000	23383.000000	23383.000000	23383.000000
mean	113.421657	1.750118	106.759749	1.153359
std	32.819372	0.809895	29.982431	0.532176
min	43.080000	1.000000	43.080000	1.000000
25%	90.000000	1.000000	85.085000	1.000000
50%	106.000000	2.000000	100.250000	1.000000
75%	129.315000	2.000000	124.950000	1.000000
max	430.940000	3.000000	376.320000	3.000000

	MFCCPreschool	MFCCPreschool_flag	_75FCCInfant	_75FCCInfant_flag \
count	23383.000000	23383.000000	23383.000000	23383.000000
mean	104.189510	1.287859	128.909289	1.792841
std	28.961701	0.696762	38.543010	0.818080

min	40.030000	1.000000	50.000000	1.000000
25%	84.255000	1.000000	100.830000	1.000000
50%	99.650000	1.000000	123.150000	2.000000
75%	120.200000	1.000000	146.950000	3.000000
max	331.340000	3.000000	502.970000	3.000000

	_75FCCToddler	_75FCCToddler_flag	_75FCCPreschool	\
count	23383.000000	23383.00000	23383.000000	
mean	120.784283	1.18800	117.897482	
std	35.334666	0.58367	34.111188	
min	50.000000	1.00000	46.450000	
25%	95.850000	1.00000	95.000000	
50%	115.000000	1.00000	112.500000	
75%	136.270000	1.00000	132.760000	
max	439.220000	3.00000	386.720000	

	_75FCCPreschool_flag
count	23383.000000
mean	1.294316
std	0.708542
min	1.000000
25%	1.000000
50%	1.000000
75%	1.000000
max	3.000000

```
[15]: # More detailed statistics including additional percentiles
df_clean.describe(percentiles=[.1, .25, .5, .75, .9, .95, .99])
```

```
[15]:
```

	County_FIPS_Code	StudyYear	UNR_16	FUNR_16	\
count	34567.000000	34567.000000	34567.000000	34567.00000	
mean	30388.132786	2012.999711	7.465902	7.02902	
std	15161.015383	3.162232	3.538619	3.56342	
min	1001.000000	2008.000000	0.000000	0.00000	
10%	9013.000000	2009.000000	3.400000	2.98000	
25%	18177.000000	2010.000000	5.100000	4.64000	
50%	29177.000000	2013.000000	7.050000	6.59000	
75%	45081.000000	2016.000000	9.350000	8.88000	
90%	51017.000000	2017.000000	11.870000	11.58000	
95%	53063.000000	2018.000000	13.760000	13.54000	
99%	55125.000000	2018.000000	18.290400	18.00340	
max	56045.000000	2018.000000	36.110000	38.24000	

	MUNR_16	UNR_20to64	FUNR_20to64	MUNR_20to64	FLFPR_20to64	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	7.860291	6.900073	6.482007	7.275457	70.086125	
std	4.037657	3.446199	3.477956	3.990758	7.696499	

min	0.000000	0.000000	0.000000	0.000000	33.600000
10%	3.300000	3.000000	2.600000	2.800000	60.300000
25%	5.200000	4.600000	4.200000	4.700000	65.100000
50%	7.390000	6.500000	6.000000	6.800000	70.600000
75%	9.920000	8.700000	8.250000	9.200000	75.500000
90%	12.760000	11.200000	10.900000	12.100000	79.600000
95%	14.940000	13.000000	12.800000	14.300000	81.900000
99%	20.906800	17.634000	17.217000	20.600000	85.400000
max	39.740000	33.900000	44.500000	45.500000	100.000000

	FLFPR_20to64_Under6	FLFPR_20to64_6to17	FLFPR_20to64_Under6_6to17	\
count	34567.000000	34567.000000	34567.000000	
mean	68.821409	78.824106	66.940759	
std	11.758088	8.529813	13.285061	
min	0.000000	0.000000	0.000000	
10%	54.896000	68.500000	50.870000	
25%	62.600000	74.400000	59.800000	
50%	69.600000	79.450000	67.660000	
75%	76.100000	84.225000	75.200000	
90%	82.200000	88.654000	82.400000	
95%	86.200000	91.280000	87.117000	
99%	95.810200	96.990000	100.000000	
max	100.000000	100.000000	100.000000	

	MLFPR_20to64	PR_F	PR_P	MHI	ME	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	78.637814	11.739125	16.038131	46432.001931	27075.054081	
std	10.802983	5.681003	6.511816	12457.050093	5483.484930	
min	8.200000	0.000000	0.000000	18528.560000	4417.000000	
10%	65.400000	5.600000	8.600000	33150.200000	21311.000000	
25%	74.200000	7.700000	11.400000	38250.500000	23636.000000	
50%	81.000000	10.900000	15.200000	44438.000000	26366.000000	
75%	85.900000	14.600000	19.500000	52078.390000	30022.500000	
90%	89.300000	19.000000	24.500000	61564.800000	33374.800000	
95%	90.900000	22.100000	27.800000	70022.000000	36703.800000	
99%	94.000000	30.700000	36.300000	88262.000000	44605.360000	
max	100.000000	52.100000	55.100000	136268.000000	70565.000000	

	FME	MME	MHI_2018	ME_2018	FME_2018	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	21625.047495	33134.232570	50446.777363	29404.295539	23475.169710	
std	4716.600817	7240.296013	13279.833788	5715.192737	4903.836942	
min	4290.000000	3784.000000	19841.920000	4947.040000	5019.300000	
10%	16497.000000	25509.200000	36325.380000	23456.332000	18270.840000	
25%	18721.000000	28729.000000	41839.215000	25955.055000	20613.100000	
50%	21089.000000	32059.000000	48505.600000	28653.920000	22854.700000	
75%	23919.000000	36543.500000	56201.450000	32005.820000	25634.665000	

90%	27233.400000	41772.400000	66214.086000	35974.804000	29348.190000
95%	30269.000000	46214.000000	75861.747000	39729.487000	32363.562000
99%	36456.680000	55170.160000	95169.893200	48320.209400	39151.774800
max	62919.000000	93000.000000	136268.000000	70565.000000	68600.000000

	MME_2018	TotalPop	OneRace	OneRace_W	OneRace_B \
count	34567.000000	3.456700e+04	34567.000000	34567.000000	34567.000000
mean	35997.433918	9.914222e+04	97.922863	83.636247	8.971213
std	7643.855532	3.177786e+05	1.952523	16.683515	14.510260
min	4238.080000	4.100000e+01	59.500000	3.100000	0.000000
10%	27861.676000	5.130000e+03	96.300000	60.200000	0.200000
25%	31447.505000	1.101450e+04	97.500000	77.000000	0.600000
50%	35103.000000	2.571100e+04	98.400000	90.200000	2.100000
75%	39503.335000	6.670350e+04	98.900000	95.600000	10.200000
90%	44854.748000	1.994624e+05	99.400000	97.400000	30.200000
95%	49823.004000	4.257017e+05	99.600000	98.000000	42.000000
99%	59913.536400	1.223946e+06	100.000000	99.000000	64.300000
max	98580.000000	1.010572e+07	100.000000	100.000000	87.400000

	OneRace_I	OneRace_A	OneRace_H	OneRace_Other	TwoRaces \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	1.895634	1.216209	0.082101	2.117914	2.077146
std	7.517786	2.633768	0.593602	3.779858	1.952520
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.030000	0.000000	0.000000	0.100000	0.600000
25%	0.200000	0.200000	0.000000	0.300000	1.100000
50%	0.300000	0.500000	0.000000	0.800000	1.600000
75%	0.800000	1.200000	0.100000	2.200000	2.500000
90%	2.600000	2.700000	0.200000	5.400000	3.700000
95%	7.000000	4.400000	0.300000	8.700000	4.800000
99%	41.900000	12.900000	0.900000	19.100000	10.100000
max	95.700000	52.200000	35.300000	58.800000	40.500000

	Hispanic	Households	H_Under6_BothWork	H_Under6_FWork \
count	34567.000000	3.456700e+04	34567.000000	34567.000000
mean	8.401054	3.686299e+04	2791.610004	1821.628808
std	13.284933	1.116470e+05	8709.085062	6740.389344
min	0.000000	2.200000e+01	0.000000	0.000000
10%	1.000000	1.983600e+03	111.000000	56.000000
25%	1.700000	4.236000e+03	260.000000	146.000000
50%	3.400000	9.814000e+03	647.000000	387.000000
75%	8.400000	2.549350e+04	1786.000000	1099.000000
90%	21.700000	7.516440e+04	5757.400000	3417.400000
95%	37.300000	1.556513e+05	12358.400000	7581.900000
99%	67.400000	4.726167e+05	34049.540000	23165.660000
max	99.200000	3.306109e+06	264439.000000	246269.000000

	H_Under6_MWork	H_Under6_SingleM	H_6to17_BothWork	H_6to17_Fwork	\
count	34567.000000	34565.000000	34567.000000	34567.000000	
mean	141.109064	1992.249964	6480.568548	2874.374374	
std	472.557239	7009.934446	19294.344762	10901.264277	
min	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	59.000000	314.000000	97.000000	
25%	5.000000	176.000000	685.000000	247.000000	
50%	31.000000	489.000000	1662.000000	620.000000	
75%	103.000000	1298.000000	4469.500000	1767.000000	
90%	290.000000	3826.200000	13483.400000	5419.400000	
95%	595.000000	8091.800000	28778.600000	11920.700000	
99%	1747.340000	25562.880000	76763.980000	38979.940000	
max	15396.000000	220530.000000	590730.000000	391225.000000	

	H_6to17_Mwork	H_6to17_SingleM	EMP_M	MEMP_M	\
count	34567.000000	34565.000000	34567.000000	34567.000000	
mean	390.113056	3924.511558	30.720191	26.495400	
std	1187.096922	13966.245194	6.503783	8.013214	
min	0.000000	0.000000	0.000000	0.000000	
10%	9.000000	124.000000	23.650000	17.780000	
25%	39.000000	351.000000	26.340000	20.945000	
50%	116.000000	907.000000	29.690000	25.190000	
75%	306.000000	2421.000000	34.100000	30.720000	
90%	798.000000	7542.000000	39.200000	37.100000	
95%	1573.000000	16118.600000	42.870000	41.540000	
99%	4537.020000	50704.400000	50.900200	51.076800	
max	40499.000000	436538.000000	74.030000	76.890000	

	FEMP_M	EMP_Service	MEMP_Service	FEMP_Service	EMP_Sales	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	35.509491	17.909716	13.122111	23.424303	22.244044	
std	6.073988	3.602868	4.476415	4.696024	3.352621	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
10%	28.640000	13.910000	7.970000	17.990000	18.050000	
25%	31.560000	15.590000	10.270000	20.410000	20.150000	
50%	34.940000	17.580000	12.840000	23.090000	22.410000	
75%	38.870000	19.780000	15.450000	26.010000	24.540000	
90%	43.230000	22.340000	18.420000	29.180000	26.290000	
95%	46.197000	24.150000	20.760000	31.437000	27.290000	
99%	52.886800	29.173400	26.923400	37.063400	29.370000	
max	100.000000	46.420000	56.330000	62.780000	50.770000	

	MEMP_Sales	FEMP_Sales	EMP_N	MEMP_N	FEMP_N	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	13.660359	32.062036	13.076182	23.265887	1.369745	
std	3.761803	4.480591	4.257660	6.763236	1.430796	
min	0.000000	0.000000	0.000000	0.000000	0.000000	

10%	8.950000	26.640000	8.250000	15.270000	0.300000
25%	11.210000	29.390000	10.200000	18.680000	0.590000
50%	13.680000	32.170000	12.560000	22.700000	0.990000
75%	16.240000	34.800000	15.340000	27.210000	1.650000
90%	18.510000	37.310000	18.450000	32.010000	2.740000
95%	19.670000	39.010000	20.680000	35.130000	3.810000
99%	21.670000	43.093400	25.966800	42.300000	7.340000
max	49.060000	100.000000	60.000000	80.240000	27.270000

	EMP_P	MEMP_P	FEMP_P	iUNR_16	iFUNR_16 \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	16.049861	23.456238	7.634403	0.038881	0.038881
std	5.935141	7.956927	4.501044	0.193314	0.193314
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	8.790000	13.150000	3.030000	0.000000	0.000000
25%	11.640000	17.610000	4.410000	0.000000	0.000000
50%	15.590000	23.400000	6.630000	0.000000	0.000000
75%	19.830000	28.930000	9.920000	0.000000	0.000000
90%	23.940000	33.810000	13.834000	0.000000	0.000000
95%	26.557000	36.680000	16.410000	0.000000	0.000000
99%	31.280000	41.693400	21.733400	1.000000	1.000000
max	72.850000	87.500000	66.480000	1.000000	1.000000

	iMUNR_16	iUNR_20to64	iFUNR_20to64	iMUNR_20to64	iFLFPR_20to64 \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	0.038881	0.038881	0.038881	0.038881	0.038881
std	0.193314	0.193314	0.193314	0.193314	0.193314
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	0.000000	0.000000
90%	0.000000	0.000000	0.000000	0.000000	0.000000
95%	0.000000	0.000000	0.000000	0.000000	0.000000
99%	1.000000	1.000000	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

	iFLFPR_20to64_Under6	iFLFPR_20to64_6to17	iFLFPR_20to64_Under6_6to17 \
count	34567.000000	34567.000000	34567.000000
mean	0.040530	0.636590	0.637082
std	0.197201	0.480988	0.480849
min	0.000000	0.000000	0.000000
10%	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	1.000000	1.000000
75%	0.000000	1.000000	1.000000
90%	0.000000	1.000000	1.000000

95%	0.000000	1.000000	1.000000
99%	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000

	iMLFPR_20to64	iPR_F	iPR_P	iMHI	iME \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	0.038881	0.038245	0.038158	0.038245	0.038447
std	0.193314	0.191789	0.191580	0.191789	0.192276
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	0.000000	0.000000
90%	0.000000	0.000000	0.000000	0.000000	0.000000
95%	0.000000	0.000000	0.000000	0.000000	0.000000
99%	1.000000	1.000000	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

	iFME	iMME	iMHI_2018	iME_2018	iFME_2018 \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	0.038476	0.038505	0.909133	0.909191	0.909104
std	0.192345	0.192415	0.287424	0.287342	0.287465
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.000000	0.000000	1.000000	1.000000	1.000000
25%	0.000000	0.000000	1.000000	1.000000	1.000000
50%	0.000000	0.000000	1.000000	1.000000	1.000000
75%	0.000000	0.000000	1.000000	1.000000	1.000000
90%	0.000000	0.000000	1.000000	1.000000	1.000000
95%	0.000000	0.000000	1.000000	1.000000	1.000000
99%	1.000000	1.000000	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

	iMME_2018	iTotalPop	iOneRace	iOneRace_W	iOneRace_B \
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	0.90922	0.038187	0.044956	0.044956	0.044956
std	0.28730	0.191650	0.207211	0.207211	0.207211
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	1.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	0.000000	0.000000	0.000000	0.000000
50%	1.000000	0.000000	0.000000	0.000000	0.000000
75%	1.000000	0.000000	0.000000	0.000000	0.000000
90%	1.000000	0.000000	0.000000	0.000000	0.000000
95%	1.000000	0.000000	0.000000	0.000000	0.000000
99%	1.000000	1.000000	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

iOneRace_I	iOneRace_A	iOneRace_H	iOneRace_Other	iTwoRaces \
------------	------------	------------	----------------	-------------

count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000
mean	0.044956	0.044956	0.044956	0.044956	0.044956
std	0.207211	0.207211	0.207211	0.207211	0.207211
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	0.000000	0.000000
90%	0.000000	0.000000	0.000000	0.000000	0.000000
95%	0.000000	0.000000	0.000000	0.000000	0.000000
99%	1.000000	1.000000	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

	iHispanic	iHouseholds	iH_Under6_BothWork	iH_Under6_FWork	\
count	34567.000000	34567.000000	34567.000000	34567.000000	
mean	0.046808	0.038187	0.038447	0.038447	
std	0.211230	0.191650	0.192276	0.192276	
min	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.000000	0.000000	
90%	0.000000	0.000000	0.000000	0.000000	
95%	0.000000	0.000000	0.000000	0.000000	
99%	1.000000	1.000000	1.000000	1.000000	
max	1.000000	1.000000	1.000000	1.000000	

	iH_Under6_MWork	iH_Under6_SingleM	iH_6to17_BothWork	iH_6to17_Fwork	\
count	34567.000000	34567.000000	34567.000000	34567.000000	
mean	0.038447	0.040732	0.038447	0.038447	
std	0.192276	0.197673	0.192276	0.192276	
min	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.000000	0.000000	
90%	0.000000	0.000000	0.000000	0.000000	
95%	0.000000	0.000000	0.000000	0.000000	
99%	1.000000	1.000000	1.000000	1.000000	
max	1.000000	1.000000	1.000000	1.000000	

	iH_6to17_Mwork	iH_6to17_SingleM	iEMP_M	iMEMP_M	iFEMP_M	\
count	34567.000000	34567.000000	34567.0	34567.0	34567.0	
mean	0.038447	0.040732	1.0	1.0	1.0	
std	0.192276	0.197673	0.0	0.0	0.0	
min	0.000000	0.000000	1.0	1.0	1.0	
10%	0.000000	0.000000	1.0	1.0	1.0	

25%	0.000000	0.000000	1.0	1.0	1.0
50%	0.000000	0.000000	1.0	1.0	1.0
75%	0.000000	0.000000	1.0	1.0	1.0
90%	0.000000	0.000000	1.0	1.0	1.0
95%	0.000000	0.000000	1.0	1.0	1.0
99%	1.000000	1.000000	1.0	1.0	1.0
max	1.000000	1.000000	1.0	1.0	1.0

	iEMP_Service	iMEMP_Service	iFEMP_Service	iEMP_Sales	iMEMP_Sales	\
count	34567.0	34567.0	34567.0	34567.0	34567.0	
mean	1.0	1.0	1.0	1.0	1.0	
std	0.0	0.0	0.0	0.0	0.0	
min	1.0	1.0	1.0	1.0	1.0	
10%	1.0	1.0	1.0	1.0	1.0	
25%	1.0	1.0	1.0	1.0	1.0	
50%	1.0	1.0	1.0	1.0	1.0	
75%	1.0	1.0	1.0	1.0	1.0	
90%	1.0	1.0	1.0	1.0	1.0	
95%	1.0	1.0	1.0	1.0	1.0	
99%	1.0	1.0	1.0	1.0	1.0	
max	1.0	1.0	1.0	1.0	1.0	

	iFEMP_Sales	iEMP_N	iMEMP_N	iFEMP_N	iEMP_P	iMEMP_P	iFEMP_P	\
count	34567.0	34567.0	34567.0	34567.0	34567.0	34567.0	34567.0	
mean	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
std	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
min	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
10%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
25%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
50%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
75%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
90%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
95%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
99%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	
max	1.0	1.0	1.0	1.0	1.0	1.0	1.0	

	MCBto5	MC6to11	MC12to17	MC18to23	MC24to29	\
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000	
mean	145.930609	145.929977	142.769363	139.820997	130.482768	
std	53.722917	53.730359	52.498825	51.290259	43.775370	
min	27.730000	27.730000	27.730000	27.730000	21.540000	
10%	94.790000	94.790000	92.500000	90.480000	89.380000	
25%	108.450000	108.450000	107.000000	104.070000	100.000000	
50%	134.400000	134.400000	131.150000	128.540000	120.990000	
75%	166.330000	166.330000	165.000000	161.080000	148.710000	
90%	221.000000	221.000000	211.040000	207.550000	185.000000	
95%	251.966000	251.966000	249.944000	245.000000	215.000000	

99%	339.290000	339.290000	324.442400	316.754800	285.046400
max	470.000000	470.000000	470.000000	462.330000	419.000000

	MC30to35	MC36to41	MC42to47	MC48to53	MC54toSA \
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000
mean	128.298475	122.197682	122.197192	120.114848	120.115166
std	42.444909	38.528236	38.528509	37.474593	37.474395
min	21.540000	21.540000	21.540000	21.540000	21.540000
10%	86.450000	85.000000	85.000000	84.230000	84.230000
25%	98.670000	95.880000	95.880000	94.790000	94.790000
50%	120.000000	113.980000	113.980000	112.000000	112.000000
75%	147.200000	139.280000	139.280000	136.500000	136.500000
90%	181.570000	175.000000	175.000000	172.000000	172.000000
95%	211.490000	196.850000	196.850000	191.000000	191.000000
99%	270.500000	249.230000	249.230000	245.500000	245.500000
max	419.000000	385.000000	385.000000	385.000000	385.000000

	MCSA	MFCCBto5	MFCC6to11	MFCC12to17	MFCC18to23 \
count	23593.000000	23383.000000	23383.000000	23383.000000	23383.000000
mean	101.234253	113.252743	113.228623	111.982142	109.849199
std	34.552888	32.946207	32.897397	32.423459	31.923247
min	18.980000	40.040000	40.040000	43.080000	43.080000
10%	62.380000	79.250000	79.250000	78.250000	77.110000
25%	78.650000	90.000000	90.000000	89.955000	87.110000
50%	96.530000	106.000000	106.000000	105.100000	103.850000
75%	119.380000	129.250000	129.250000	128.000000	125.000000
90%	145.740000	155.304000	155.186000	152.500000	150.000000
95%	165.346000	175.000000	175.000000	170.000000	167.080000
99%	208.150000	219.460000	219.460000	215.000000	210.000000
max	375.400000	351.290000	351.290000	430.940000	430.940000

	MFCC24to29	MFCC30to35	MFCC36to41	MFCC42to47	MFCC48to53 \
count	23383.000000	23383.000000	23383.000000	23383.000000	23383.000000
mean	106.739308	106.301836	104.189913	104.163946	103.331517
std	29.970529	29.477408	28.960359	28.956597	28.837612
min	43.080000	43.080000	40.030000	40.030000	40.030000
10%	75.600000	75.510000	75.000000	75.000000	74.900000
25%	85.000000	85.060000	83.985000	83.985000	83.030000
50%	100.250000	100.000000	99.660000	99.650000	99.000000
75%	124.950000	124.500000	120.200000	120.200000	119.960000
90%	146.700000	145.000000	140.540000	140.540000	140.000000
95%	161.462000	158.964000	153.268000	153.268000	152.898000
99%	200.000000	200.000000	200.000000	200.000000	198.678400
max	376.320000	376.320000	331.340000	331.340000	331.340000

	MFCC54toSA	MFCCSA	_75CBto5	_75C6to11	_75C12to17 \
count	23383.000000	23383.000000	23593.000000	23593.000000	23593.000000

mean	103.319273	92.523582	165.747724	165.723671	162.313312
std	28.823258	27.669904	62.948905	62.923413	61.354370
min	40.030000	22.000000	32.180000	32.180000	32.180000
10%	74.900000	62.224000	107.600000	107.600000	105.000000
25%	83.030000	75.000000	123.450000	123.450000	120.180000
50%	99.000000	88.180000	150.000000	150.000000	148.130000
75%	119.960000	107.500000	185.000000	185.000000	182.600000
90%	140.000000	127.390000	253.992000	253.992000	244.064000
95%	152.898000	140.000000	294.220000	294.220000	290.608000
99%	198.678400	177.500000	400.120000	400.120000	383.414800
max	331.340000	308.000000	515.000000	515.000000	515.000000

	_75C18to23	_75C24to29	_75C30to35	_75C36to41	_75C42to47	\
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000	
mean	159.300156	149.211592	146.869878	140.109674	140.344320	
std	60.338314	52.230764	50.595279	46.157201	46.329759	
min	32.180000	25.000000	25.000000	25.000000	25.000000	
10%	104.090000	100.000000	99.230000	97.380000	97.380000	
25%	119.000000	113.580000	111.670000	108.780000	108.780000	
50%	145.000000	137.400000	135.000000	129.230000	129.380000	
75%	180.000000	169.330000	167.320000	158.680000	159.540000	
90%	238.170000	216.500000	211.814000	203.232000	204.840000	
95%	286.958000	255.000000	250.000000	230.798000	230.798000	
99%	378.150000	334.532000	318.680000	301.464400	301.464400	
max	492.210000	485.000000	485.000000	430.000000	430.000000	

	_75C48to53	_75C54toSA	_75CSA	_75FCCBto5	_75FCC6to11	\
count	23593.000000	23593.000000	23593.000000	23383.000000	23383.000000	
mean	138.134505	138.135380	118.607736	128.621556	128.646967	
std	45.395519	45.395017	40.732376	38.586132	38.599945	
min	25.000000	25.000000	22.030000	47.210000	47.210000	
10%	95.000000	95.000000	73.460000	89.482000	89.440000	
25%	106.880000	106.880000	92.130000	100.040000	100.040000	
50%	127.500000	127.500000	112.500000	122.650000	122.800000	
75%	156.650000	156.650000	138.890000	146.950000	146.950000	
90%	200.000000	200.000000	170.560000	178.866000	178.866000	
95%	226.212000	226.212000	195.000000	200.000000	200.000000	
99%	299.016800	299.016800	249.520000	254.010000	254.010000	
max	430.000000	430.000000	540.000000	410.000000	410.000000	

	_75FCC12to17	_75FCC18to23	_75FCC24to29	_75FCC30to35	_75FCC36to41	\
count	23383.000000	23383.000000	23383.000000	23383.000000	23383.000000	
mean	127.100881	124.452879	120.755554	120.081898	117.868539	
std	37.887589	37.178615	35.302723	34.840404	34.101515	
min	50.000000	50.000000	50.000000	50.000000	46.450000	
10%	88.600000	86.240000	84.000000	84.000000	83.390000	
25%	100.000000	98.710000	95.595000	95.450000	95.000000	

50%	121.460000	120.000000	115.000000	115.000000	112.500000
75%	146.000000	141.430000	136.270000	136.250000	132.760000
90%	175.000000	173.500000	168.420000	164.530000	160.728000
95%	198.180000	194.800000	185.396000	181.480000	177.730000
99%	252.311400	250.000000	236.368800	235.000000	234.653800
max	502.970000	502.970000	439.220000	439.220000	386.720000

	_75FCC42to47	_75FCC48to53	_75FCC54toSA	_75FCCSA	iMCBto5 \
count	23383.000000	23383.000000	23383.000000	23383.000000	23593.000000
mean	117.868539	117.060965	117.060965	107.514791	538.150045
std	34.101515	33.957578	33.957578	31.439791	747.570838
min	46.450000	46.450000	46.450000	39.950000	0.000000
10%	83.390000	82.500000	82.500000	75.000000	0.000000
25%	95.000000	94.225000	94.225000	86.250000	1.000000
50%	112.500000	111.670000	111.670000	100.000000	111.000000
75%	132.760000	131.540000	131.540000	125.000000	1001.000000
90%	160.728000	159.180000	159.180000	147.680000	1100.000000
95%	177.730000	176.590000	176.590000	161.094000	1101.000000
99%	234.653800	233.780000	233.780000	211.042400	1101.000000
max	386.720000	386.720000	386.720000	341.220000	11001.000000

	iMC6to11	iMC12to17	iMC18to23	iMC24to29	iMC30to35 \
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000
mean	538.150045	527.129827	527.553681	514.414191	514.414191
std	747.570838	676.514314	679.934582	584.844914	584.844914
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	1.000000	1.000000	1.000000	1.000000
50%	111.000000	111.000000	111.000000	111.000000	111.000000
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000
90%	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000
95%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000
99%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000
max	11001.000000	11001.000000	11001.000000	11000.000000	11000.000000

	iMC36to41	iMC42to47	iMC48to53	iMC54toSA	iMCSA \
count	23593.000000	23593.000000	23593.000000	23593.000000	23593.000000
mean	509.751791	509.751791	509.751791	509.751791	593.674988
std	544.014342	544.014342	544.014342	544.014342	1081.932707
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	1.000000	1.000000	1.000000	1.000000
50%	111.000000	111.000000	111.000000	111.000000	111.000000
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000
90%	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000
95%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000
99%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000

max	10001.000000	10001.000000	10001.000000	10001.000000	11101.000000	
	iMFCCBto5	iMFCC6to11	iMFCC12to17	iMFCC18to23	iMFCC24to29	\
count	23363.000000	23363.000000	23363.000000	23363.000000	23363.000000	
mean	535.735094	537.447203	529.742713	513.477678	501.492916	
std	783.397312	795.248421	745.691945	641.250142	547.793159	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	111.000000	111.000000	111.000000	111.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
90%	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000	
95%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000	
99%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000	
max	11100.000000	11100.000000	11100.000000	11100.000000	11100.000000	
	iMFCC30to35	iMFCC36to41	iMFCC42to47	iMFCC48to53	iMFCC54toSA	\
count	23363.000000	23363.000000	23363.000000	23363.000000	23363.000000	
mean	501.492916	499.352780	499.352780	498.068698	498.068698	
std	547.793159	525.786117	525.786117	512.172349	512.172349	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	111.000000	111.000000	111.000000	111.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
90%	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000	
95%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000	
99%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000	
max	11100.000000	11100.000000	11100.000000	11100.000000	11100.000000	
	iMFCCSA	i_75CBto5	i_75C6to11	i_75C12to17	i_75C18to23	\
count	23363.000000	23575.000000	23575.000000	23575.000000	23575.000000	
mean	559.704618	594.387869	594.387869	581.238346	581.662524	
std	920.348454	792.244081	792.244081	713.358381	716.573259	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	111.000000	1000.000000	1000.000000	1000.000000	1000.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
90%	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000	
95%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000	
99%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000	
max	11101.000000	11001.000000	11001.000000	11001.000000	11001.000000	
	i_75C24to29	i_75C30to35	i_75C36to41	i_75C42to47	i_75C48to53	\
count	23575.000000	23575.000000	23575.000000	23575.000000	23575.000000	
mean	564.695398	564.695398	571.482248	561.301972	561.301972	

std	594.180636	594.180636	639.646517	564.529253	564.529253
min	0.000000	0.000000	0.000000	0.000000	0.000000
10%	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	1.000000	1.000000	1.000000	1.000000
50%	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000
90%	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000
95%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000
99%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000
max	11000.000000	11000.000000	11000.000000	11000.000000	11000.000000

	i_75C54toSA	i_75CSA	i_75FCCBto5	i_75FCC6to11	i_75FCC12to17	\
count	23575.000000	23575.000000	23383.000000	23383.000000	23383.000000	
mean	561.301972	666.498155	596.021811	597.732455	590.889877	
std	564.529253	1155.682656	786.101669	797.773319	755.200166	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	1.000000	
50%	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	
90%	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000	
95%	1101.000000	1101.000000	1101.000000	1101.000000	1101.000000	
99%	1101.000000	10000.000000	1101.000000	1101.000000	1101.000000	
max	11000.000000	11100.000000	11100.000000	11100.000000	11100.000000	

	i_75FCC18to23	i_75FCC24to29	i_75FCC30to35	i_75FCC36to41	\
count	23383.000000	23383.000000	23383.000000	23383.000000	
mean	574.638755	562.236582	562.236582	566.513193	
std	653.899222	559.793463	559.793463	589.392174	
min	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	
50%	1000.000000	1000.000000	1000.000000	1000.000000	
75%	1001.000000	1001.000000	1001.000000	1001.000000	
90%	1100.000000	1100.000000	1100.000000	1100.000000	
95%	1101.000000	1101.000000	1101.000000	1101.000000	
99%	1101.000000	1101.000000	1101.000000	1101.000000	
max	11100.000000	11100.000000	11100.000000	11100.000000	

	i_75FCC42to47	i_75FCC48to53	i_75FCC54toSA	i_75FCCSA	\
count	23383.000000	23383.000000	23383.000000	23383.000000	
mean	566.513193	566.513193	566.513193	663.164607	
std	589.392174	589.392174	589.392174	1101.122681	
min	0.000000	0.000000	0.000000	0.000000	
10%	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	1.000000	1.000000	1.000000	
50%	1000.000000	1000.000000	1000.000000	1000.000000	

75%	1001.000000	1001.000000	1001.000000	1001.000000
90%	1100.000000	1100.000000	1100.000000	1100.000000
95%	1101.000000	1101.000000	1101.000000	1101.000000
99%	1101.000000	1101.000000	1101.000000	10000.000000
max	11100.000000	11100.000000	11100.000000	11100.000000

	MCInfant	MCInfant_flag	MCToddler	MCToddler_flag \
count	23593.000000	23593.000000	23593.000000	23593.000000
mean	146.051770	1.899801	130.482768	1.315263
std	53.698566	0.831546	43.775370	0.728806
min	27.730000	1.000000	21.540000	1.000000
10%	95.000000	1.000000	89.380000	1.000000
25%	108.750000	1.000000	100.000000	1.000000
50%	134.500000	2.000000	120.990000	1.000000
75%	166.330000	3.000000	148.710000	1.000000
90%	221.070000	3.000000	185.000000	3.000000
95%	251.966000	3.000000	215.000000	3.000000
99%	339.290000	3.000000	285.046400	3.000000
max	470.000000	3.000000	419.000000	3.000000

	MCPreschool	MCPreschool_flag	_75CInfant	_75CInfant_flag \
count	23593.000000	23593.000000	23593.000000	23593.000000
mean	122.232852	1.367609	165.926607	1.926461
std	38.538323	0.774638	62.889701	0.837951
min	21.540000	1.000000	32.180000	1.000000
10%	85.000000	1.000000	107.950000	1.000000
25%	95.880000	1.000000	123.700000	1.000000
50%	113.990000	1.000000	150.000000	2.000000
75%	139.300000	1.000000	185.000000	3.000000
90%	175.000000	3.000000	254.230000	3.000000
95%	196.850000	3.000000	294.220000	3.000000
99%	249.230000	3.000000	400.120000	3.000000
max	385.000000	3.000000	515.000000	3.000000

	_75CToddler	_75CToddler_flag	_75CPreschool	_75CPreschool_flag \
count	23593.000000	23593.000000	23593.000000	23593.000000
mean	149.213983	1.307634	140.386184	1.370661
std	52.236542	0.721561	46.337349	0.774716
min	25.000000	1.000000	25.000000	1.000000
10%	100.000000	1.000000	97.380000	1.000000
25%	113.580000	1.000000	108.900000	1.000000
50%	137.400000	1.000000	129.450000	1.000000
75%	169.330000	1.000000	159.580000	1.000000
90%	216.500000	3.000000	204.918000	3.000000
95%	255.000000	3.000000	230.798000	3.000000
99%	334.670000	3.000000	301.464400	3.000000
max	485.000000	3.000000	430.000000	3.000000

	MFCCInfant	MFCCInfant_flag	MFCCToddler	MFCCToddler_flag \
count	23383.000000	23383.000000	23383.000000	23383.000000
mean	113.421657	1.750118	106.759749	1.153359
std	32.819372	0.809895	29.982431	0.532176
min	43.080000	1.000000	43.080000	1.000000
10%	79.400000	1.000000	75.600000	1.000000
25%	90.000000	1.000000	85.085000	1.000000
50%	106.000000	2.000000	100.250000	1.000000
75%	129.315000	2.000000	124.950000	1.000000
90%	155.432000	3.000000	146.700000	1.000000
95%	175.000000	3.000000	161.479000	3.000000
99%	219.460000	3.000000	200.000000	3.000000
max	430.940000	3.000000	376.320000	3.000000

	MFCCPreschool	MFCCPreschool_flag	_75FCCInfant	_75FCCInfant_flag \
count	23383.000000	23383.000000	23383.000000	23383.000000
mean	104.189510	1.287859	128.909289	1.792841
std	28.961701	0.696762	38.543010	0.818080
min	40.030000	1.000000	50.000000	1.000000
10%	75.000000	1.000000	89.850000	1.000000
25%	84.255000	1.000000	100.830000	1.000000
50%	99.650000	1.000000	123.150000	2.000000
75%	120.200000	1.000000	146.950000	3.000000
90%	140.540000	3.000000	178.952000	3.000000
95%	153.290000	3.000000	201.275000	3.000000
99%	200.000000	3.000000	254.010000	3.000000
max	331.340000	3.000000	502.970000	3.000000

	_75FCCToddler	_75FCCToddler_flag	_75FCCPreschool \
count	23383.000000	23383.000000	23383.000000
mean	120.784283	1.18800	117.897482
std	35.334666	0.58367	34.111188
min	50.000000	1.00000	46.450000
10%	84.000000	1.00000	83.390000
25%	95.850000	1.00000	95.000000
50%	115.000000	1.00000	112.500000
75%	136.270000	1.00000	132.760000
90%	168.420000	1.00000	160.780000
95%	185.400000	3.00000	177.730000
99%	236.368800	3.00000	234.991800
max	439.220000	3.00000	386.720000

	_75FCCPreschool_flag
count	23383.000000
mean	1.294316
std	0.708542

```

min                1.000000
10%                1.000000
25%                1.000000
50%                1.000000
75%                1.000000
90%                3.000000
95%                3.000000
99%                3.000000
max                3.000000

```

```

[16]: # Identify potential price-related columns
price_columns = [col for col in df_clean.columns if any(keyword in col.lower()
    for keyword in ['price', 'cost', 'fee', 'mhi', 'wage', 'income'])]

print(f"Price-related columns ({len(price_columns)}):")
for col in price_columns:
    print(f"    - {col}")

```

Price-related columns (4):

```

- MHI
- MHI_2018
- iMHI
- iMHI_2018

```

```

[17]: # Summary statistics for price-related columns
if price_columns:
    price_stats = df_clean[price_columns].describe()
    print("Summary Statistics for Price-Related Columns:")
    print("="*60)
    print(price_stats)

```

Summary Statistics for Price-Related Columns:

```

=====

```

	MHI	MHI_2018	iMHI	iMHI_2018
count	34567.000000	34567.000000	34567.000000	34567.000000
mean	46432.001931	50446.777363	0.038245	0.909133
std	12457.050093	13279.833788	0.191789	0.287424
min	18528.560000	19841.920000	0.000000	0.000000
25%	38250.500000	41839.215000	0.000000	1.000000
50%	44438.000000	48505.600000	0.000000	1.000000
75%	52078.390000	56201.450000	0.000000	1.000000
max	136268.000000	136268.000000	1.000000	1.000000

```

[18]: # Calculate additional statistics for price columns
if price_columns:
    for col in price_columns[:5]: # First 5 price columns
        print(f"\n{col}:")

```



```

print(f" Mean: ${df_clean[col].mean():,.2f}")
print(f" Median: ${df_clean[col].median():,.2f}")
print(f" Std Dev: ${df_clean[col].std():,.2f}")
print(f" Min: ${df_clean[col].min():,.2f}")
print(f" Max: ${df_clean[col].max():,.2f}")
print(f" Non-null count: {df_clean[col].count():,}")

```

MHI:

```

Mean: $46,432.00
Median: $44,438.00
Std Dev: $12,457.05
Min: $18,528.56
Max: $136,268.00
Non-null count: 34,567

```

MHI_2018:

```

Mean: $50,446.78
Median: $48,505.60
Std Dev: $13,279.83
Min: $19,841.92
Max: $136,268.00
Non-null count: 34,567

```

iMHI:

```

Mean: $0.04
Median: $0.00
Std Dev: $0.19
Min: $0.00
Max: $1.00
Non-null count: 34,567

```

iMHI_2018:

```

Mean: $0.91
Median: $1.00
Std Dev: $0.29
Min: $0.00
Max: $1.00
Non-null count: 34,567

```

1.4 4. Geographic Analysis of Childcare Prices

Analyze childcare prices across different geographic regions.

```

[19]: # Identify geographic columns
geo_columns = [col for col in df_clean.columns if any(keyword in col.lower()
               for keyword in ['state', 'county', 'region', 'city'])]

```

```
print(f"Geographic columns ({len(geo_columns)}):")
for col in geo_columns:
    print(f"    - {col}")
```

Geographic columns (4):

- State_Name
- State_Abbreviation
- County_Name
- County_FIPS_Code

```
[20]: # Analyze by state (if state column exists)
state_col = next((col for col in df_clean.columns if 'state' in col.lower()),
                 ↪None)

if state_col:
    print(f"Number of unique states: {df_clean[state_col].nunique()}")
    print(f"\nStates in dataset:")
    print(df_clean[state_col].value_counts().head(20))
```

Number of unique states: 51

States in dataset:

State_Name

Texas	2794
Georgia	1749
Virginia	1469
Kentucky	1320
Missouri	1265
Kansas	1155
Illinois	1122
North Carolina	1100
Iowa	1089
Tennessee	1045
Nebraska	1023
Indiana	1012
Ohio	968
Minnesota	957
Michigan	913
Mississippi	902
Oklahoma	847
Arkansas	825
Wisconsin	792
Pennsylvania	737

Name: count, dtype: int64

```
[21]: # Calculate average prices by state for different childcare types
if state_col and price_columns:
    # Select a few key price columns for state-level analysis
```

```

key_price_cols = price_columns[:5]

state_prices = df_clean.groupby(state_col)[key_price_cols].mean().round(2)
state_prices = state_prices.sort_values(by=key_price_cols[0],
↪ascending=False)

print("Average Prices by State (Top 15):")
print("="*60)
print(state_prices.head(15))

```

Average Prices by State (Top 15):

```

=====

```

	MHI	MHI_2018	iMHI	iMHI_2018
State_Name				
New Jersey	73233.99	79623.80	0.00	0.91
Connecticut	71303.62	77540.83	0.00	0.91
Maryland	69173.71	75202.56	0.00	0.91
District of Columbia	66970.55	72509.74	0.00	0.91
Massachusetts	66082.41	71796.33	0.01	0.91
Rhode Island	65828.09	71562.28	0.00	0.91
Hawaii	63846.18	69352.66	0.02	0.91
Alaska	62591.14	67993.62	0.07	0.91
New Hampshire	60200.88	65407.13	0.00	0.91
Delaware	57959.91	63004.42	0.00	0.91
California	57110.38	62078.51	0.01	0.91
Wyoming	55704.23	60532.59	0.05	0.91
New York	54861.69	59615.46	0.00	0.91
Utah	54178.00	58831.37	0.05	0.91
Nevada	53817.06	58504.36	0.05	0.91

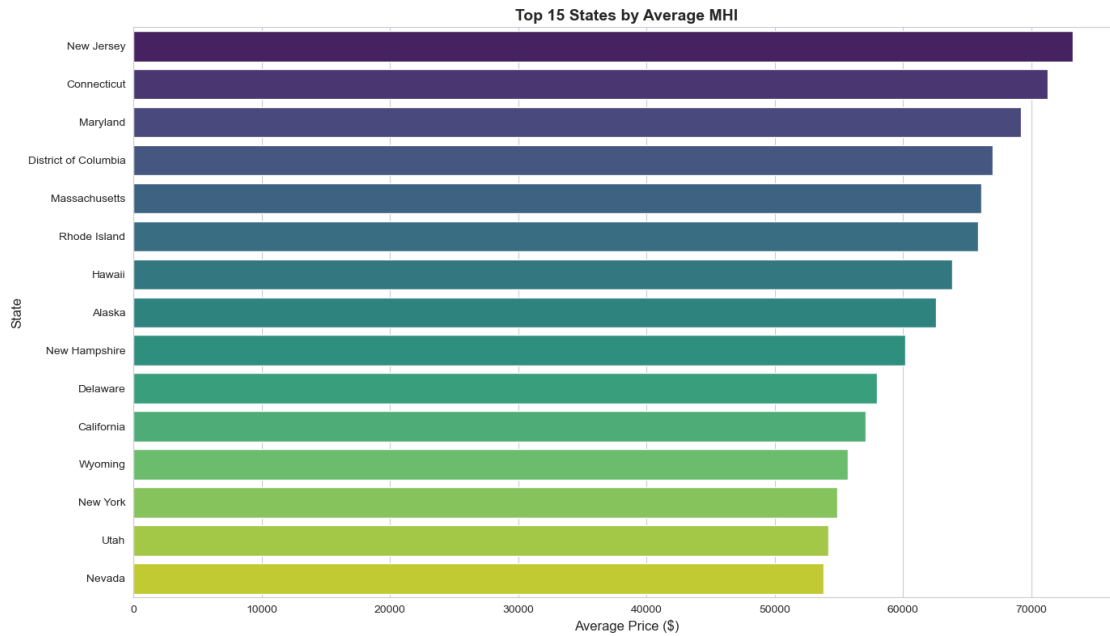
```

[22]: # Visualize top 15 states by childcare price (using first price column)
if state_col and price_columns:
    plt.figure(figsize=(14, 8))

    state_avg = df_clean.groupby(state_col)[price_columns[0]].mean().
↪sort_values(ascending=False).head(15)

    sns.barplot(x=state_avg.values, y=state_avg.index, palette='viridis')
    plt.xlabel('Average Price ($)', fontsize=12)
    plt.ylabel('State', fontsize=12)
    plt.title(f'Top 15 States by Average {price_columns[0]}', fontsize=14,
↪fontweight='bold')
    plt.tight_layout()
    plt.show()

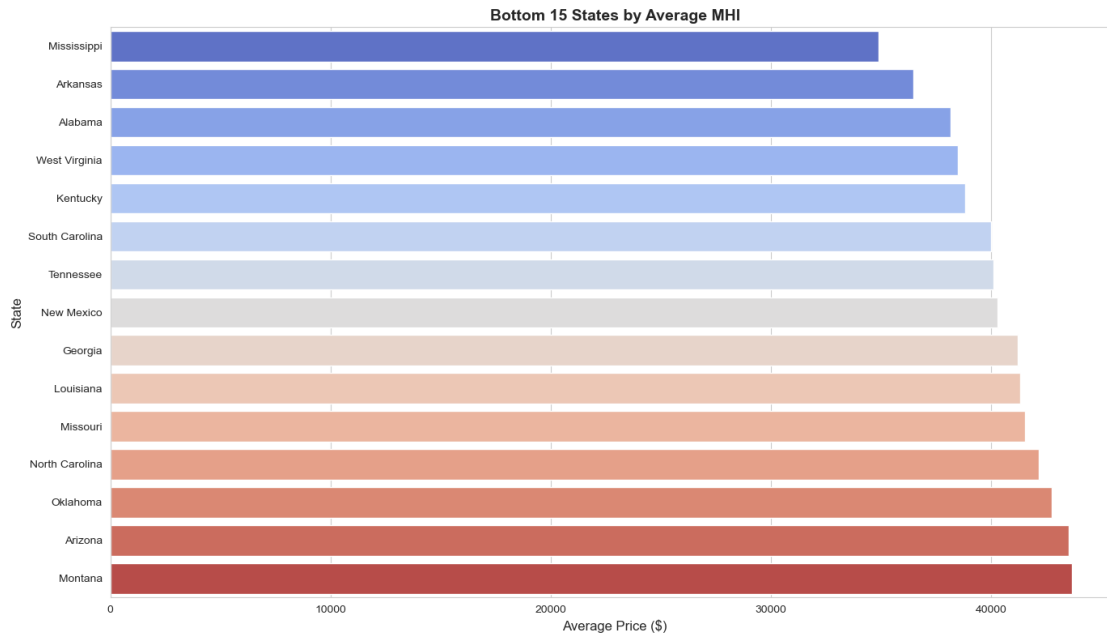
```



```
[23]: # Visualize bottom 15 states by childcare price
if state_col and price_columns:
    plt.figure(figsize=(14, 8))

    state_avg_low = df_clean.groupby(state_col)[price_columns[0]].mean().
    ↪sort_values(ascending=True).head(15)

    sns.barplot(x=state_avg_low.values, y=state_avg_low.index,
    ↪palette='coolwarm')
    plt.xlabel('Average Price ($)', fontsize=12)
    plt.ylabel('State', fontsize=12)
    plt.title(f'Bottom 15 States by Average {price_columns[0]}', fontsize=14,
    ↪fontweight='bold')
    plt.tight_layout()
    plt.show()
```



```
[24]: # Geographic price variation analysis
if state_col and price_columns:
    geo_analysis = df_clean.groupby(state_col)[price_columns[0]].agg(['mean', '
    ↪ 'median', 'std', 'min', 'max'])
    geo_analysis = geo_analysis.sort_values('mean', ascending=False)

    print("Geographic Price Variation (Top 10 States):")
    print("="*60)
    print(geo_analysis.head(10).round(2))
```

Geographic Price Variation (Top 10 States):

```
=====
```

	mean	median	std	min	max
State_Name					
New Jersey	73233.99	71374.0	16302.87	49537.00	112535.0
Connecticut	71303.62	69787.5	8707.06	57427.00	92969.0
Maryland	69173.71	66587.5	20297.63	35154.00	117730.0
District of Columbia	66970.55	65830.0	8607.24	56428.00	82604.0
Massachusetts	66082.41	64187.5	13069.38	47513.00	105171.0
Rhode Island	65828.09	69240.0	9014.50	48500.00	81301.0
Hawaii	63846.18	64058.0	8498.17	41806.00	82906.0
Alaska	62591.14	62830.5	12999.39	33154.89	90749.0
New Hampshire	60200.88	58496.0	10436.49	41087.00	90429.0
Delaware	57959.91	55184.0	5845.04	49794.00	70996.0

1.5 5. Price Comparison Across Different Age Groups

Compare childcare prices for different age categories.

```
[25]: # Identify age-related price columns
age_price_columns = [col for col in price_columns if any(keyword in col.lower()
    for keyword in ['infant', 'toddler', 'preschool',
    ↪ 'school', 'age'])]

print(f"Age-related price columns ({len(age_price_columns)}):")
for col in age_price_columns:
    print(f" - {col}")
```

Age-related price columns (0):

```
[26]: # Compare average prices across age groups
if age_price_columns:
    age_comparison = df_clean[age_price_columns].mean().
    ↪ sort_values(ascending=False)

    print("Average Prices by Age Group:")
    print("="*60)
    for col, price in age_comparison.items():
        print(f"{col}: ${price:,.2f}")
```

```
[27]: # Visualize price comparison across age groups
if age_price_columns and len(age_price_columns) > 0:
    plt.figure(figsize=(12, 6))

    age_means = df_clean[age_price_columns].mean().sort_values(ascending=False)

    sns.barplot(x=age_means.index, y=age_means.values, palette='Set2')
    plt.xticks(rotation=45, ha='right')
    plt.ylabel('Average Price ($)', fontsize=12)
    plt.xlabel('Age Group Category', fontsize=12)
    plt.title('Average Childcare Prices by Age Group', fontsize=14,
    ↪ fontweight='bold')
    plt.tight_layout()
    plt.show()
```

```
[28]: # Box plot comparison across age groups
if age_price_columns and len(age_price_columns) > 0:
    plt.figure(figsize=(14, 7))

    # Prepare data for box plot
    age_data = df_clean[age_price_columns].melt(var_name='Age_Group',
    ↪ value_name='Price')
    age_data = age_data.dropna()
```

```

sns.boxplot(data=age_data, x='Age_Group', y='Price', palette='Set3')
plt.xticks(rotation=45, ha='right')
plt.ylabel('Price ($)', fontsize=12)
plt.xlabel('Age Group', fontsize=12)
plt.title('Distribution of Childcare Prices by Age Group', fontsize=14,
fontweight='bold')
plt.tight_layout()
plt.show()

```

```

[29]: # Statistical comparison of age groups
if age_price_columns:
    print("Detailed Statistics by Age Group:")
    print("="*60)
    age_stats = df_clean[age_price_columns].describe()
    print(age_stats.round(2))

```

1.6 6. Time Series Analysis of Price Trends

Analyze how childcare prices have changed over time (if temporal data is available).

```

[30]: # Identify time-related columns
time_columns = [col for col in df_clean.columns if any(keyword in col.lower()
for keyword in ['year', 'date', 'time', 'period'])]

print(f"Time-related columns ({len(time_columns)}):")
for col in time_columns:
    print(f" - {col}")
    print(f" Unique values: {df_clean[col].nunique()}")
    if df_clean[col].nunique() <= 20:
        print(f" Values: {sorted(df_clean[col].unique())}")

```

```

Time-related columns (1):
- StudyYear
Unique values: 11
Values: [2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018]

```

```

[31]: # Time series analysis if year column exists
year_col = next((col for col in time_columns if 'year' in col.lower()), None)

if year_col and price_columns:
    yearly_trends = df_clean.groupby(year_col)[price_columns[:3]].mean()

    print("Average Prices by Year:")
    print("="*60)
    print(yearly_trends.round(2))

```

Average Prices by Year:

=====

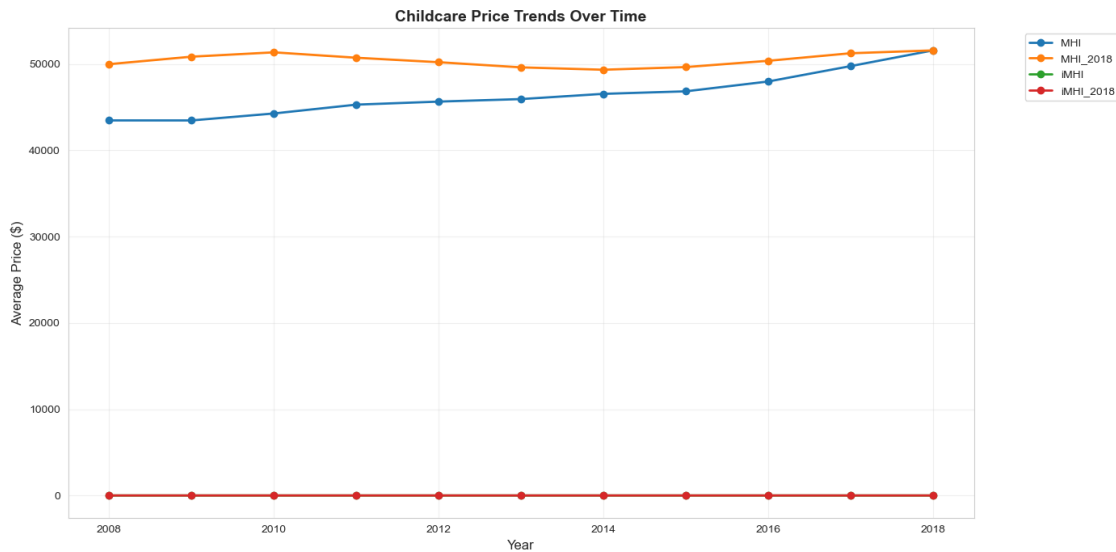
	MHI	MHI_2018	iMHI
StudyYear			
2008	43464.60	49984.29	0.42
2009	43462.07	50850.63	0.00
2010	44270.30	51353.55	0.00
2011	45292.07	50727.12	0.00
2012	45644.41	50208.86	0.00
2013	45937.12	49612.09	0.00
2014	46543.76	49336.39	0.00
2015	46834.98	49645.08	0.00
2016	47973.23	50371.89	0.00
2017	49754.16	51246.79	0.00
2018	51577.71	51577.71	0.00

```
[32]: # Visualize price trends over time
if year_col and price_columns:
    plt.figure(figsize=(14, 7))

    yearly_avg = df_clean.groupby(year_col)[price_columns[:5]].mean()

    for col in price_columns[:5]:
        plt.plot(yearly_avg.index, yearly_avg[col], marker='o', label=col,
        linewidth=2)

    plt.xlabel('Year', fontsize=12)
    plt.ylabel('Average Price ($)', fontsize=12)
    plt.title('Childcare Price Trends Over Time', fontsize=14,
    fontweight='bold')
    plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()
```

```
[33]: # Calculate year-over-year growth rates
if year_col and price_columns:
    yearly_prices = df_clean.groupby(year_col)[price_columns[0]].mean()
    yoy_growth = yearly_prices.pct_change() * 100

    print("Year-over-Year Growth Rates (%):")
    print("="*60)
    print(yoy_growth.round(2))
```

Year-over-Year Growth Rates (%):

```
=====
StudyYear
2008      NaN
2009    -0.01
2010     1.86
2011     2.31
2012     0.78
2013     0.64
2014     1.32
2015     0.63
2016     2.43
2017     3.71
2018     3.67
Name: MHI, dtype: float64
```

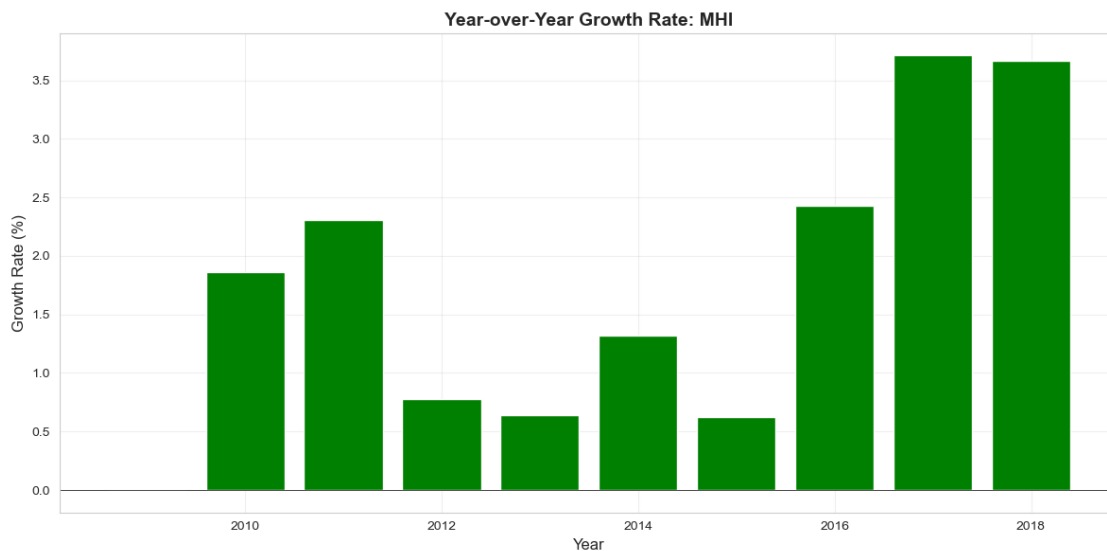
```
[34]: # Visualize year-over-year growth
if year_col and price_columns:
    plt.figure(figsize=(12, 6))
```

```

yearly_prices = df_clean.groupby(year_col)[price_columns[0]].mean()
yoy_growth = yearly_prices.pct_change() * 100

plt.bar(yoy_growth.index, yoy_growth.values, color=['green' if x > 0 else
↪'red' for x in yoy_growth.values])
plt.xlabel('Year', fontsize=12)
plt.ylabel('Growth Rate (%)', fontsize=12)
plt.title(f'Year-over-Year Growth Rate: {price_columns[0]}', fontsize=14,
↪fontweight='bold')
plt.axhline(y=0, color='black', linestyle='-', linewidth=0.5)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```



1.7 7. Correlation Analysis

Examine relationships between childcare prices and other variables.

```

[35]: # Calculate correlation matrix for numeric columns
correlation_matrix = df_clean[numeric_columns].corr()

print("Correlation matrix shape:", correlation_matrix.shape)

```

Correlation matrix shape: (224, 224)

```

[36]: # Display correlations with price columns
if price_columns:
    price_correlations = correlation_matrix[price_columns[0]].
↪sort_values(ascending=False)

```

```

print(f"Correlations with {price_columns[0]}:")
print("="*60)
print(price_correlations.head(20))

```

Correlations with MHI:

```

=====
MHI                1.000000
MHI_2018           0.981759
ME                 0.840663
ME_2018           0.826697
MME               0.809142
MME_2018          0.786898
FME               0.738467
FME_2018          0.726215
MC48to53          0.657259
MC54toSA          0.657248
MCPreschool       0.643447
MC42to47          0.642864
MC36to41          0.642847
_75C54toSA        0.638956
_75C48to53        0.638951
MC12to17          0.638601
MFCC12to17        0.637738
MC30to35          0.637037
_75C36to41        0.633905
_75CPreschool     0.631346
Name: MHI, dtype: float64

```

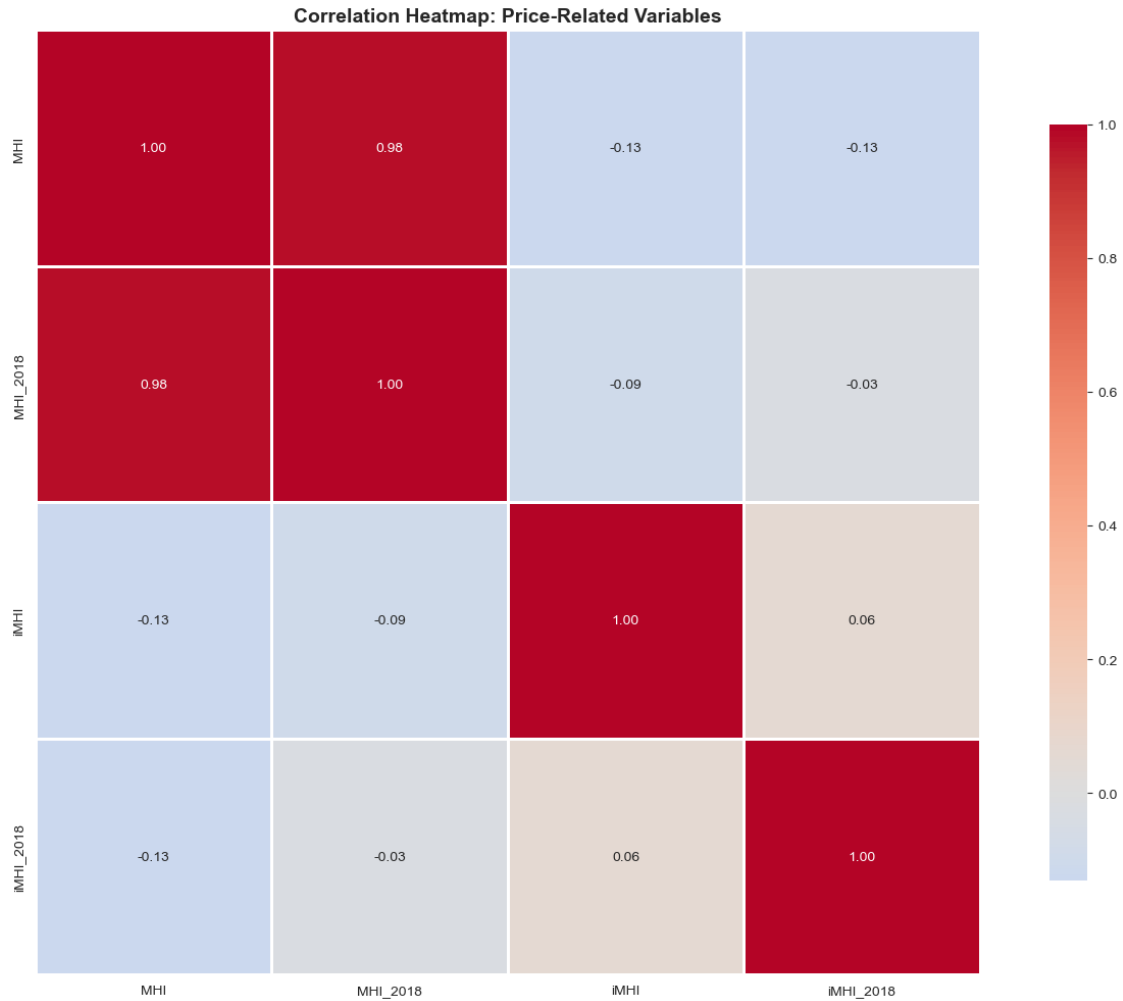
```

[37]: # Visualize correlation heatmap for price-related columns
if len(price_columns) > 1:
    plt.figure(figsize=(14, 10))

    # Select price columns and highly correlated variables
    cols_for_heatmap = price_columns[:10] if len(price_columns) > 10 else price_columns
    corr_subset = df_clean[cols_for_heatmap].corr()

    sns.heatmap(corr_subset, annot=True, fmt='.2f', cmap='coolwarm',
                center=0, square=True, linewidths=1, cbar_kws={"shrink": 0.8})
    plt.title('Correlation Heatmap: Price-Related Variables', fontsize=14,
             fontweight='bold')
    plt.tight_layout()
    plt.show()

```



```
[38]: # Full correlation heatmap (for all numeric variables)
plt.figure(figsize=(20, 16))

sns.heatmap(correlation_matrix, cmap='coolwarm', center=0,
            square=True, linewidths=0.5, cbar_kws={"shrink": 0.8})
plt.title('Full Correlation Heatmap: All Numeric Variables', fontsize=16,
        fontweight='bold')
plt.tight_layout()
plt.show()
```



```

key=abs,
ascending=False)

print(f"Strong Correlations (|r| > 0.7): {len(strong_corr_df)}")
print("="*60)
print(strong_corr_df.head(20))

```

Strong Correlations (|r| > 0.7): 2420

```

=====

```

	Variable 1	Variable 2	Correlation
532	iOneRace_W	iOneRace_B	1.0
655	iH_Under6_MWork	iH_6to17_Fwork	1.0
2204	iMC42to47	iMC48to53	1.0
2197	iMC36to41	iMC54toSA	1.0
2196	iMC36to41	iMC48to53	1.0
133	iFUNR_16	iMUNR_16	1.0
2195	iMC36to41	iMC42to47	1.0
2170	iMC24to29	iMC30to35	1.0
665	iH_6to17_Fwork	iH_6to17_Mwork	1.0
663	iH_6to17_BothWork	iH_6to17_Mwork	1.0
662	iH_6to17_BothWork	iH_6to17_Fwork	1.0
661	iH_Under6_SingleM	iH_6to17_SingleM	1.0
656	iH_Under6_MWork	iH_6to17_Mwork	1.0
654	iH_Under6_MWork	iH_6to17_BothWork	1.0
922	MC24to29	MCToddler	1.0
651	iH_Under6_FWork	iH_6to17_Mwork	1.0
650	iH_Under6_FWork	iH_6to17_Fwork	1.0
101	iUNR_16	iMUNR_16	1.0
100	iUNR_16	iFUNR_16	1.0
2142	iMCBto5	iMC6to11	1.0

```

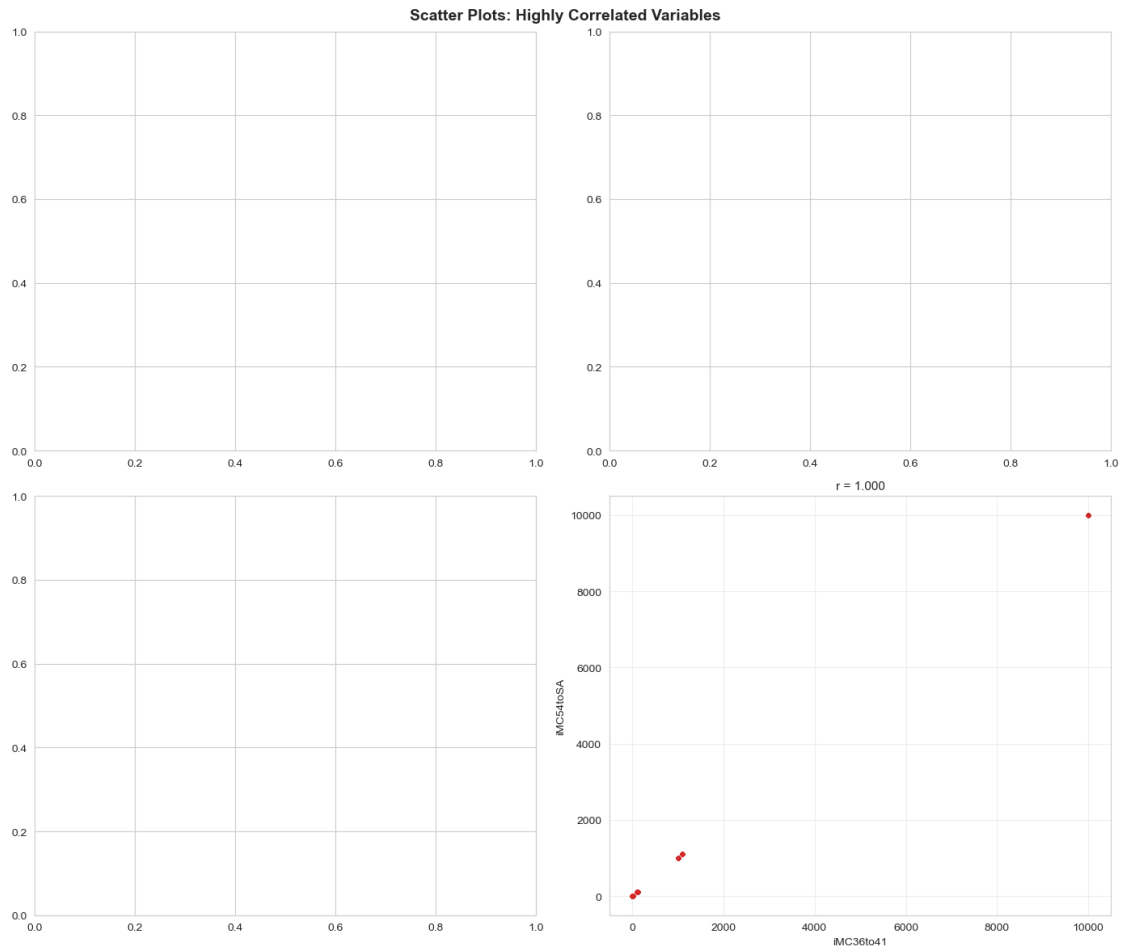
[40]: # Scatter plot for highly correlated variables
if len(strong_corr_df) > 0 and len(strong_corr_df) > 0:
    fig, axes = plt.subplots(2, 2, figsize=(14, 12))
    axes = axes.ravel()

    for idx, row in strong_corr_df.head(4).iterrows():
        var1, var2 = row['Variable 1'], row['Variable 2']

        ax = axes[idx if idx < 4 else 3]
        ax.scatter(df_clean[var1], df_clean[var2], alpha=0.5, s=10)
        ax.set_xlabel(var1, fontsize=10)
        ax.set_ylabel(var2, fontsize=10)
        ax.set_title(f'r = {row["Correlation"]:.3f}', fontsize=11)
        ax.grid(True, alpha=0.3)

```

```
plt.suptitle('Scatter Plots: Highly Correlated Variables', fontsize=14,
↪fontweight='bold')
plt.tight_layout()
plt.show()
```



1.8 8. Visualization of Key Findings

Comprehensive visualizations to communicate main insights.

```
[41]: # Create a comprehensive dashboard visualization
if price_columns and state_col:
    fig, axes = plt.subplots(2, 2, figsize=(16, 12))

    # 1. Distribution of main price variable
    axes[0, 0].hist(df_clean[price_columns[0]].dropna(), bins=50,
↪color='skyblue', edgecolor='black')
    axes[0, 0].set_xlabel('Price ($)', fontsize=11)
```

```

axes[0, 0].set_ylabel('Frequency', fontsize=11)
axes[0, 0].set_title(f'Distribution of {price_columns[0]}', fontsize=12,
fontweight='bold')
axes[0, 0].grid(True, alpha=0.3)

# 2. Top 10 states by average price
state_avg = df_clean.groupby(state_col)[price_columns[0]].mean().
sort_values(ascending=False).head(10)
axes[0, 1].barh(range(len(state_avg)), state_avg.values, color='coral')
axes[0, 1].set_yticks(range(len(state_avg)))
axes[0, 1].set_yticklabels(state_avg.index)
axes[0, 1].set_xlabel('Average Price ($)', fontsize=11)
axes[0, 1].set_title('Top 10 States by Average Price', fontsize=12,
fontweight='bold')
axes[0, 1].grid(True, alpha=0.3, axis='x')

# 3. Price comparison across age groups (if available)
if age_price_columns and len(age_price_columns) > 0:
    age_means = df_clean[age_price_columns[:5]].mean().
    sort_values(ascending=False)
    axes[1, 0].bar(range(len(age_means)), age_means.values,
color='lightgreen', edgecolor='black')
    axes[1, 0].set_xticks(range(len(age_means)))
    axes[1, 0].set_xticklabels([col[:20] for col in age_means.index],
rotation=45, ha='right', fontsize=9)
    axes[1, 0].set_ylabel('Average Price ($)', fontsize=11)
    axes[1, 0].set_title('Average Prices by Age Group', fontsize=12,
fontweight='bold')
    axes[1, 0].grid(True, alpha=0.3, axis='y')

# 4. Box plot of price distribution
price_data = df_clean[price_columns[:5]].melt(var_name='Category',
value_name='Price')
price_data = price_data.dropna()

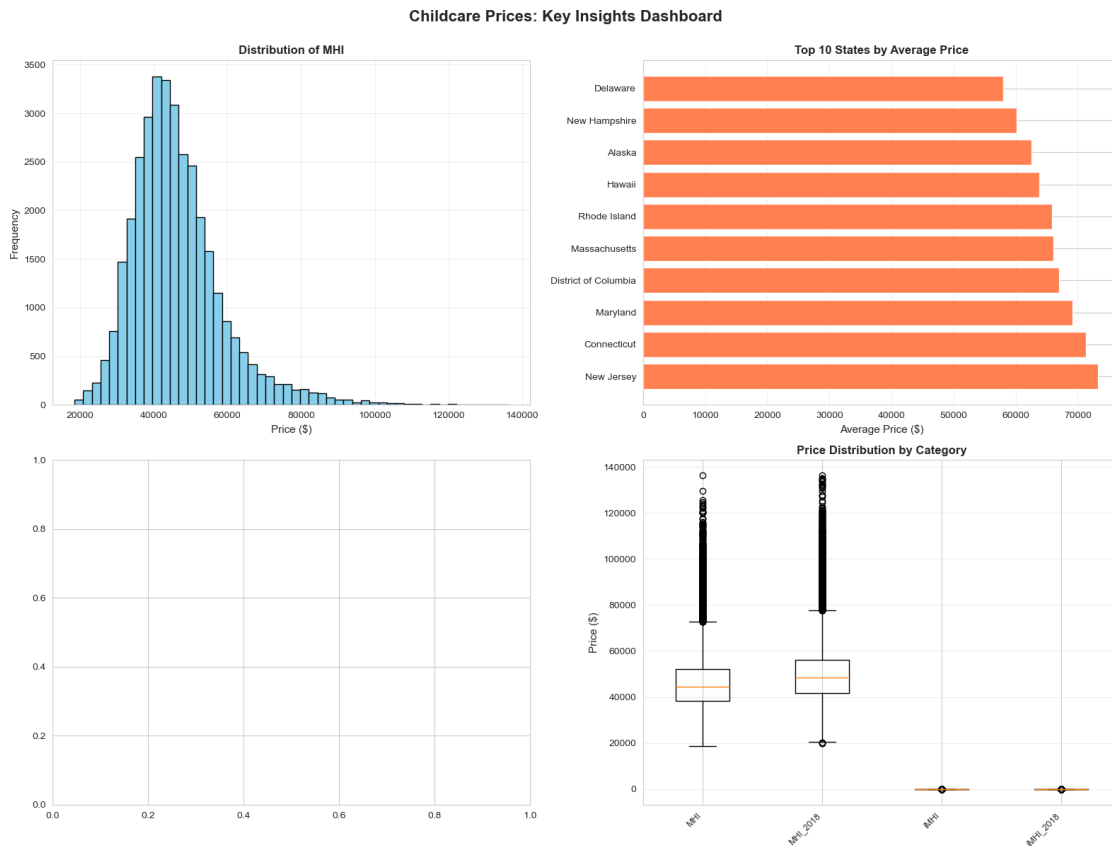
categories = price_data['Category'].unique()[:5]
box_data = [price_data[price_data['Category'] == cat]['Price'].values for
cat in categories]

axes[1, 1].boxplot(box_data, labels=[cat[:15] for cat in categories])
axes[1, 1].set_xticklabels([cat[:15] for cat in categories], rotation=45,
ha='right', fontsize=9)
axes[1, 1].set_ylabel('Price ($)', fontsize=11)
axes[1, 1].set_title('Price Distribution by Category', fontsize=12,
fontweight='bold')
axes[1, 1].grid(True, alpha=0.3, axis='y')

```



```
plt.suptitle('Childcare Prices: Key Insights Dashboard', fontsize=16,
fontweight='bold', y=1.00)
plt.tight_layout()
plt.show()
```



```
[42]: # Summary statistics visualization
if price_columns:
    fig, axes = plt.subplots(1, 2, figsize=(16, 6))

    # Price range comparison
    price_ranges = df_clean[price_columns[:8]].agg(['min', 'max', 'mean'])
    x_pos = np.arange(len(price_ranges.columns))

    axes[0].bar(x_pos - 0.2, price_ranges.loc['min'], width=0.2, label='Min',
color='lightblue')
    axes[0].bar(x_pos, price_ranges.loc['mean'], width=0.2, label='Mean',
color='orange')
    axes[0].bar(x_pos + 0.2, price_ranges.loc['max'], width=0.2, label='Max',
color='lightcoral')
```

```

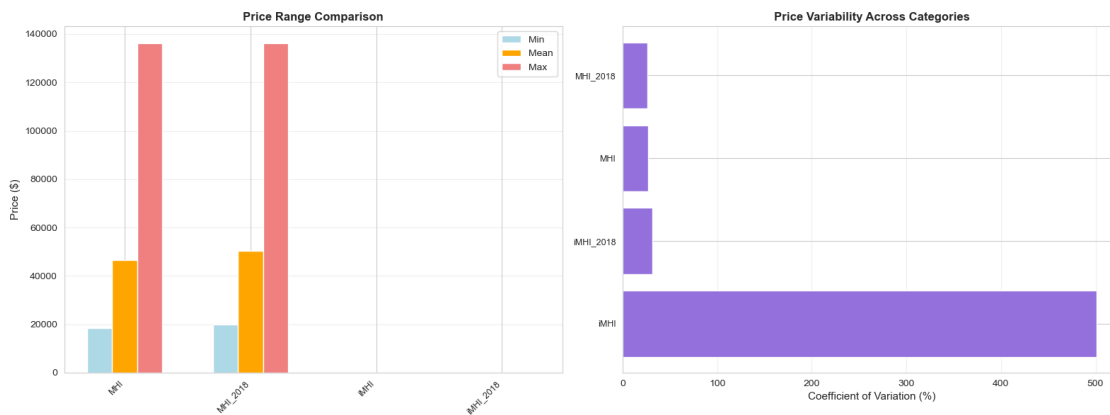
axes[0].set_xticks(x_pos)
axes[0].set_xticklabels([col[:20] for col in price_ranges.columns],
↪rotation=45, ha='right', fontsize=9)
axes[0].set_ylabel('Price ($)', fontsize=11)
axes[0].set_title('Price Range Comparison', fontsize=12, fontweight='bold')
axes[0].legend()
axes[0].grid(True, alpha=0.3, axis='y')

# Coefficient of variation (to show relative variability)
cv_data = (df_clean[price_columns[:8]].std() / df_clean[price_columns[:8]].
↪mean()) * 100
cv_data = cv_data.sort_values(ascending=False)

axes[1].barh(range(len(cv_data)), cv_data.values, color='mediumpurple')
axes[1].set_yticks(range(len(cv_data)))
axes[1].set_yticklabels([col[:25] for col in cv_data.index], fontsize=9)
axes[1].set_xlabel('Coefficient of Variation (%)', fontsize=11)
axes[1].set_title('Price Variability Across Categories', fontsize=12,
↪fontweight='bold')
axes[1].grid(True, alpha=0.3, axis='x')

plt.tight_layout()
plt.show()

```



```

[43]: # Geographic heatmap-style visualization
if state_col and price_columns:
    state_summary = df_clean.groupby(state_col)[price_columns[:5]].mean()

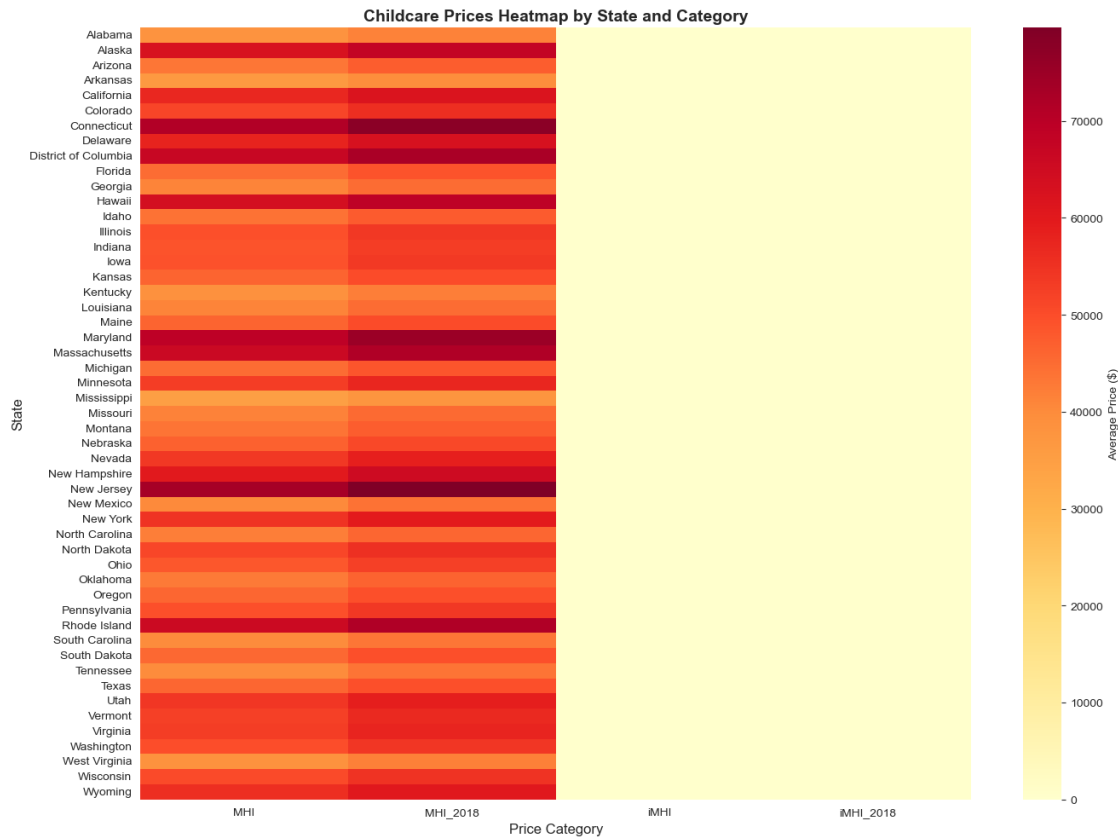
    plt.figure(figsize=(14, 10))
    sns.heatmap(state_summary, annot=False, cmap='YlOrRd', cbar_kws={'label':
↪'Average Price ($)'})

```

```

plt.title('Childcare Prices Heatmap by State and Category', fontsize=14,
fontweight='bold')
plt.xlabel('Price Category', fontsize=12)
plt.ylabel('State', fontsize=12)
plt.tight_layout()
plt.show()

```



```

[44]: # Final summary report
print("="*80)
print(" " * 20 + "CHILDCARE PRICES ANALYSIS - SUMMARY REPORT")
print("="*80)

print(f"\nDataset Overview:")
print(f" • Total Records: {len(df_clean):,}")
print(f" • Total Variables: {len(df_clean.columns)}")
print(f" • Numeric Variables: {len(numeric_columns)}")
print(f" • Price-Related Variables: {len(price_columns)}")

if state_col:
    print(f"\nGeographic Coverage:")

```

```

    print(f"    • Number of States: {df_clean[state_col].nunique()}")

if year_col:
    print(f"\nTemporal Coverage:")
    print(f"    • Years: {df_clean[year_col].min()} - {df_clean[year_col].max()}")

if price_columns:
    print(f"\nPrice Statistics ({price_columns[0]}):")
    print(f"    • Average: ${df_clean[price_columns[0]].mean():,.2f}")
    print(f"    • Median: ${df_clean[price_columns[0]].median():,.2f}")
    print(f"    • Std Dev: ${df_clean[price_columns[0]].std():,.2f}")
    print(f"    • Range: ${df_clean[price_columns[0]].min():,.2f} - ␣
↪${df_clean[price_columns[0]].max():,.2f}")

print("\n" + "="*80)
print("Analysis Complete!")
print("="*80)

```

CHILD CARE PRICES ANALYSIS - SUMMARY REPORT

Dataset Overview:

- Total Records: 34,567
- Total Variables: 227
- Numeric Variables: 224
- Price-Related Variables: 4

Geographic Coverage:

- Number of States: 51

Temporal Coverage:

- Years: 2008 - 2018

Price Statistics (MHI):

- Average: \$46,432.00
- Median: \$44,438.00
- Std Dev: \$12,457.05
- Range: \$18,528.56 - \$136,268.00

Analysis Complete!

1.9 Conclusions and Next Steps

This analysis has provided comprehensive insights into:

1. **Geographic Variations:** Significant price differences across states and regions
2. **Age Group Patterns:** Price variations based on child age categories
3. **Temporal Trends:** Changes in childcare costs over time
4. **Correlations:** Relationships between prices and demographic/economic factors

1.9.1 Recommendations for Further Analysis:

- Investigate factors driving high prices in certain states
- Analyze accessibility and affordability metrics
- Study the relationship between childcare costs and family income levels
- Examine urban vs. rural price differences
- Conduct predictive modeling for future price trends

[]: